

Identifying Drivers of Internal Displacement in Mali

A Panel Regression Analysis with Random Forest Validation

Overview

We model monthly changes in log-IDP counts across Mali's 50 admin2 cercles (2014–2024) as a function of three candidate driver families: **conflict**, **climate**, and **food prices**. Our goal is to identify the drivers of displacement — their **form, lag, and heterogeneity** — and to state **how credibly each can be read as causal**, being explicit about what a panel regression can and cannot establish.

The analysis proceeds in two parts:

1. **Model selection** (Stages A–E): a structured funnel that narrows 282 candidate features to the 11 retained in the final model (M5), growing the specification from 1 feature to the final model one step at a time.
2. **Validation and extensions** (Parts II–IV): Random Forest stress-tests the linear model, heterogeneity tests reveal what the average effects hide, and the independent refugee-outflow outcome cross-validates the findings.

Identification — what the design can claim. We have no instrument or natural experiment for the drivers, so these are not clean causal point estimates. But the design supports a **graded causal reading**, strongest where it matters most:

- **Two-way fixed effects** remove every time-invariant cercle difference and every common time shock — the dominant confounders.
- **Drivers enter only at lags** (conflict 7–12m, climate 13–24m), with the contemporaneous window excluded, so reverse causality is implausible.
- **Climate** (rainfall, SPI, streamflow) is plausibly **exogenous** to local displacement — weather is not caused by who fled — so the climate channel approaches a natural experiment.
- The headline result, **conflict spillover** (neighbour violence), cannot be reverse-caused: a cercle's own displacement cannot have produced its neighbours' earlier violence.
- **Weakest:** *local* conflict (not exogenous) and *food* (a near-national series with little within-cercle variation) — reported with corresponding caution.

We therefore identify the drivers and their structure with **graded causal confidence**, not uniform causal certainty.

Data. IDP source: DTM survey panel, zero-run-filtered and re-stitched. Sample: $n = 1,417$ observations, 44 cercles, after requiring complete data for all M5 features. Conflict: ACLED strict (direct lethal violence). Climate: ERA5 reanalysis. Food: WFP price monitoring. Spatial weights: road-distance decay matrix.

Survey structure — irregular rounds and zero-change intervals. DTM runs 81 irregular survey rounds (2014–2024) with non-uniform gaps, and many cercles miss rounds; in the raw panel **~65% of adjacent-round changes are exactly zero** (repeated figures, not fresh surveys). We apply a **months-based zero-run filter**: a run of consecutive zero-change rounds is kept when the non-zero changes bracketing it are **≤5 months apart**, but a longer run is dropped and the series re-stitched into a single survivor-to-survivor interval (3,078 → 1,622 intervals, zeros 65% → 33%, gaps now 1–96 months). The outcome is then expressed **per month** ($\div \text{months_between}$) so the re-stitched intervals stay comparable, and intervals longer than 6 months are excluded — leaving **$n = 1,417$** across 44 cercles.

Notation.

- y_{it} = monthly change in log-IDP (**y_change_per_month**) for cercle i at survey date t
- C_{it} = controls: **log1p_idp_t1** (AR anchor), **months_between**
- α_i, γ_t = cercle and time fixed effects; SE clustered on cercle
- $f(\cdot)$ = feature transform: **log1p** for non-negative counts, identity for signed anomalies
- Wx = spatial-spillover twin of x (neighbour-weighted average at the same lag band)

Three criteria for variable inclusion (applied throughout):

1. **$\Delta R^2_{\text{within}}$** — does the variable add explanatory power?
2. **p-value** — is the signal distinguishable from noise?
3. **Coefficient sign** — is the direction mechanistically coherent?

No variable earns its place on one criterion alone.

Feature construction

Outcome. The regression target is **y_change_per_month** = $(\log1p(\text{IDP}_{t2}) - \log1p(\text{IDP}_{t1})) / \text{months_between}$ — the average monthly log-IDP growth over each survey interval. Dividing by **months_between** makes observations comparable regardless of survey gap length (intervals range from 1 to 6 months).

Transforms. The transform for each feature is chosen by value distribution:

- **Conflict** (**fatalities**, **events**): always non-negative counts → **log1p** transform. This compresses the heavy right tail (rare extreme events) and makes the relationship approximately linear on a proportional scale.
- **Climate**: all 112 climate local features use **identity (no transform)**. Most are signed anomalies (SPI, temperature, rainfall, streamflow, NDVI, soil moisture) that can be negative, so log1p is not applicable. The SPI-derived duration and severity variables (**spi6_flood_duration**, **spi6_drought_severity**, etc.) are non-negative, but are kept on identity via an explicit override — these are already-derived intensity measures where log1p would distort the natural scale and reduce $|t|$ empirically.
- **Food**: **inflation_food_price_index** (a growth rate, signed) → identity. **c_food_price_index** and **c_maize** (price levels, non-negative) → **log1p**.

Lag structure — banded windows. Rather than point lags (the value at exactly N months ago), we use **non-overlapping banded window averages**. A band $[\text{lo}, \text{hi}]_m$ takes the mean of the monthly variable over the window from $\text{month_of}(t1) - \text{hi}$ to $\text{month_of}(t1) - \text{lo}$, strictly before the observation start

t1. For example:

Band	window	interpretation
1–3m	months 1, 2, 3 before t1	short-term pre-survey
7–9m	months 7, 8, 9 before t1	medium-term
10–12m	months 10, 11, 12 before t1	~one year lag
13–18m	months 13–18 before t1	slow-burn effects
19–24m	months 19–24 before t1	long-term

Banded windows are **non-overlapping** — the 1–3m and 4–6m bands contain different months, so they can be included jointly without multicollinearity from overlap. This is why distributed-lag models (including all bands simultaneously) remain feasible up to the point where between-band correlations become a problem.

Conflict uses bands 1–3m, 4–6m, 7–9m, 10–12m (no longer lags — conflict is not expected to have displacement effects at 2+ year horizons). Climate and food use the same four short bands plus 13–18m and 19–24m (slow-burn pathways exist for climate). The **current window** (0 months lag, i.e. the observation period (t1, t2] itself) is used only for climate and food — conflict in the current window is excluded because it risks reverse causality (displacement events may cause conflict rather than vice versa).

This exclusion is **by design, not by fit**. Added to M5, current-window local conflict is in fact positive, significant, and strong (+0.078**, ΔR² = +0.012 — *larger* than the lagged 7–9m term). That is precisely the problem: a strong *contemporaneous* conflict–displacement association within the same interval is exactly what reverse causality would produce (displacement can trigger or coincide with violence — retaliation, competition in receiving areas, militia activity around camps), and within one window the two directions cannot be separated. The lagged design (local 7–9m, spillover 10–12m) is what makes the causal reading defensible: past violence can precede later displacement, but later displacement cannot have caused past violence. Current conflict is excluded not because it fails, but because it *succeeds in a way we cannot interpret*.

Matching monthly predictors to the irregular intervals. Conflict, climate, and food are monthly series, while the IDP intervals are irregular and — after re-stitching — of varying length. They are aligned by calendar month, by **anchoring every lag to the interval start t1**:

- A **lagged band** [lo, hi]m is the mean of the monthly series over the calendar months month(t1)–hi ... month(t1)–lo, strictly before t1. Because it depends on t1 alone (not on t2 or the gap length), every interval — short or long, re-stitched or not — receives the same well-defined pre-interval history. Irregular gaps therefore do not distort the lagged predictors.
- The **current window** (lag 0, climate and food only) is the mean over the interval itself, (t1, t2], so it automatically spans the actual gap — averaging one month for a 1-month interval and, say, six for a re-stitched 6-month one. Pairing this with the per-month outcome keeps long and short intervals comparable.
- **Conflict has no current window** (excluded above), so it enters only through pre-t1 bands and is wholly unaffected by interval length.

So a re-stitched long interval changes only its *own*-window average (and the outcome, which is per-month normalised); the conflict and lagged-climate history that does most of the explanatory work is anchored to t1 and unchanged.

Cumulative windows. Cumulative features (e.g. total fatalities over the past 12 months) were considered in early exploratory work but **not used in the final analysis**. They overlap with adjacent banded windows, creating multicollinearity, and the banded design already captures sustained exposure via multiple non-overlapping bands. The final feature set contains banded windows only.

Feature count. The full candidate set has **282 features**: 16 conflict (2 indicators × 4 bands × local + spillover), 224 climate (16 variables × 7 lag windows × local + spillover), and 42 food (3 variables × 7 lag windows × local + spillover). The indicators are:

Family	Indicators	Transform	Lag windows
Conflict	<i>fatalities, events</i> (ACLED strict)	log1p	bands 1–3, 4–6, 7–9, 10–12m
Climate	<i>spi6, spi6_flood_dur, spi6_flood_sev, spi6_drought_dur, spi6_drought_sev, p_anom_1m, p_anom_3m, t2m_anom_1m, t2m_max_anom_1m, ndvi_anom_1m, ndvi_annual_change, ndvi_calendar_month_change, sm1_anom_1m, sm3_anom_1m, sm4_anom_1m, streamflow_anom_1m</i>	identity (signed anomalies)	current + bands 1–3, 4–6, 7–9, 10–12, 13–18, 19–24m
Food	<i>c_maize, c_food_price_index, inflation_food_price_index</i>	identity or log1p (level vs rate)	current + bands 1–3, 4–6, 7–9, 10–12, 13–18, 19–24m

Each local feature has a **spatial-spillover twin** W·X constructed as the row-standardised, zero-diagonal product of the road-distance weight matrix and the panel of local values, pivoted to (time × unit) and stacked back to long form. The spillover represents the neighbour-weighted average of the same feature at the same lag — capturing how the cercle's neighbourhood context affects local displacement.

Part I — Linear Panel Regression

Stage A — Screen: which features have any individual signal?

We fit each of the 282 candidate features alone (with controls and FE) and rank by |t|–statistic, applying a Benjamini–Hochberg correction at q < 0.10:

$$y_{it} = \beta f(x_{it}^{(k)}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}, \quad k = 1, \dots, 282$$

Result. Four features pass — all conflict spillovers. Zero climate, zero food pass individually:

Rank	Feature	coef	t-stat	q(BH)
1	$W \cdot \text{fatalities}_{7_9m}$	+0.362	3.42	0.082 ✓
2	$W \cdot \text{events}_{7_9m}$	+0.762	3.40	0.082 ✓
3	$W \cdot \text{fatalities}_{10_12m}$	+0.302	3.33	0.082 ✓
4	$W \cdot \text{fatalities}_{1_3m}$	+0.281	3.25	0.084 ✓
5	$W \cdot \text{events}_{4_6m}$	+0.674	3.11	0.102 ×
6	$\text{sm1_anom}_{1m_1_3m}$ (best climate)	+6.39	3.00	0.102 ×
7	spi6_{19_24m} (best SPI, tied)	−0.088	3.00	0.102 ×

The first four are all conflict **spillovers** ($W \cdot X$) — violence in neighbouring cercles predicts local IDP changes. Climate just misses the correction (two features tied at $q = 0.102$). Food appears nowhere near the top.

Although climate doesn't survive the individual screen, ranks 6–21 are almost entirely SPI-6 features at 13–18m and 19–24m lags — a coherent cluster just below the cutoff. This tells us climate has real signal that is simply weaker than conflict individually, and that the lag windows worth testing are 13–18m and 19–24m.

Stage B — Form: local or spillover?

The screen shows conflict spillovers dominate, but doesn't tell us whether conflict and climate should enter as **local** terms (x , the cercle's own value) or **spillover** terms (Wx , neighbours' value). We test this with the smallest joint model — one conflict + one climate/food partner, each in exactly one form:

$$y_{it} = \beta f(\text{conflict form}) + \theta f(\text{partner form}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Fitting all four local/spillover combinations across 1,064 (conflict × partner) pairs:

2-term model	median R^2_{within}	wins (of 1,064 pairs)
local x + local z	0.0615	184
spillover Wx + local z	0.0732	564
local x + spillover Wz	0.0589	68
spillover Wx + spillover Wz	0.0714	248

"Wins" means: for each pair, which of the four combinations has the highest R^2 ? $Wx + z$ wins 564 of the 1,064 head-to-head contests. The two spillover-conflict combinations together win 812 (76%).

Conclusion from Stage B. Conflict belongs as a **spillover** term; its partner belongs as a **local** term. This is the one question Stage B can answer: *in the minimal 2-term model*, which form of each driver carries the signal. It cannot tell us whether climate spillovers matter once conflict is fully controlled — that requires a larger joint model. We carry both local and spillover forms forward into Stages C and D, where the full picture emerges.

Stage C — Variable: which conflict anchor?

Stage B established the form. Stage C asks which conflict **variable** (which indicator and lag band) anchors the model. We now carry both forms of both drivers — the four-term pair — and sweep all 8 conflict × 133 climate/food combinations (1,064 pairs):

$$y_{it} = \beta_1 f(X) + \beta_2 f(WX) + \theta_1 f(Z) + \theta_2 f(WZ) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Ranking by full-model R^2 , the top five pairs are:

Rank	Conflict (X)	Partner (Z)	R^2_{full}	local coef (p)	W-X coef (p)
1	$\text{fatalities}_{10_12m}$	$\text{ndvi_anom}_{19_24m}$	0.098	+0.035 (0.074)	+0.292 (0.002)
2	$\text{fatalities}_{10_12m}$	$\text{ndvi_annual_change}_{19_24m}$	0.095	+0.036 (0.066)	+0.296 (0.001)
3	$\text{fatalities}_{10_12m}$	spi6_{13_18m}	0.095	+0.036 (0.072)	+0.304 (0.001)
4	$\text{fatalities}_{10_12m}$	t2m_anom_{7_9m}	0.094	+0.037 (0.064)	+0.309 (0.001)
5	$\text{fatalities}_{10_12m}$	sm1_anom_{4_6m}	0.094	+0.036 (0.069)	+0.304 (0.001)
550	events_{7_9m} (best events pair)	$\text{ndvi_anom}_{19_24m}$	0.069	+0.103 (0.016)	+0.727 (0.001)

Three things are immediately clear from the table. First, **$\text{fatalities}_{10_12m}$ appears in every top pair** — whatever climate partner it is matched with, the conflict anchor is always the same variable (19 of 20 top pairs). Second, the **W-X spillover coefficient is the stable, dominant signal** (+0.29–0.31*** across all top pairs), consistent with Stage B's finding that conflict acts through neighbours. Third, local conflict is also significant ($p \approx 0.06$ –0.07) but smaller — both forms are real.

The last row illustrates why W-X coefficient alone is not a sufficient criterion: events_{7_9m} has a much larger spillover coefficient (+0.727***), yet its R^2_{full} is only 0.069 vs 0.094–0.098. This is a **measurement-scale artefact** — one event is a smaller shock than $\text{fatalities}_{10_12m}$ (+0.29–0.31***),

one log-fatality, so the coefficient must be larger to produce the same displacement effect. R^2_{within} is scale-invariant and correctly identifies `fatalities_10_12m` as the better anchor.

No food partner rivals conflict. Best food-partner $R^2_{\text{full}} = 0.089$ vs climate-partner best = 0.098; only 7% of food partners are significant in the full spec.

Stage D — Exhaustive search: lock the backbone, optimal lags, and find M1

Stages B–C suggested `fatalities` as the conflict variable and indicated food adds nothing. To avoid hand-picking, we confirm this **exhaustively**. We also remove an unjustified assumption from the simpler pairwise searches: there is no reason the local and spillover conflict terms must use the same lag band. Local conflict captures violence in the focal circle itself — people flee and register quickly (~7–9m). Spillover conflict captures arrivals from neighbouring circles — a slower pipeline through DTM registration (~10–12m). We search over **independent** local lag ℓ and spillover lag s :

$$y_{it} = \underbrace{\beta_1 f(\text{conf}_\ell) + \beta_2 f(W \cdot \text{conf}_s)}_{\text{conflict}} + \underbrace{\theta_1 f(z_1) + \theta_2 f(W z_1) + \theta_3 f(z_2) + \theta_4 f(W z_2)}_{\text{two climate variables}} + \underbrace{\phi_1 f(\text{food}) + \phi_2 f(W \cdot \text{food})}_{\text{food}} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Why two climate variables? One slot forces a single pathway. Climate may act through distinct mechanisms at different lags (e.g. wet anomaly at 13–18m vs dry anomaly at 19–24m). Two slots from different physical source series let the search find the best-fitting independent pair.

3,951,360 models (2 conflict types × 4 local lags × 4 spillover lags × 21 food × 5,880 climate pairs), run on a compute cluster (~10h on 16 cores). The top models by R^2_{within} :

Rank	Type	local lag	spill lag	Food	z1	z2	R ² _within	n_sig
1	fatalities	7–9m	10–12m	infl_food_1_3m	spi6_13_18m	spi6_19_24m	0.1025	3
2	fatalities	7–9m	10–12m	infl_food_1_3m	t2m_7_9m	spi6_19_24m	0.1023	3
3	fatalities	7–9m	10–12m	infl_food_1_3m	spi6_13_18m	spi6_19_24m	0.1022	3
4	fatalities	7–9m	10–12m	infl_food_4_6m	spi6_13_18m	spi6_19_24m	0.1020	3
8	fatalities	4–6m	10–12m	infl_food_1_3m	spi6_13_18m	spi6_19_24m	0.1017	2

n_sig = regressors significant at p<0.10 (out of 8).

Three outcomes read from the table:

Outcome 1 — Conflict anchor and lags confirmed. `fatalities` is in every top model. The spillover lag is **10–12m in 100% of the top 100 models** regardless of local-lag choice — the DTM registration delay for arrivals is consistently ~10–12m. The local optimal lag is **7–9m** — people fleeing their own circle register 2–3 months faster. The mixed-lag spec (local 7–9m, spillover 10–12m) beats the same-lag (10–12m/10–12m) by +0.0027 R^2 .

Why not a short 1–3m conflict lag? A natural intuition is that conflict should drive displacement *fast* — people flee within weeks of violence, so the strongest signal should sit at 1–3m. We tested this directly, ranking every local lag band:

Local conflict lag	$\Delta R^2_{\text{within}}$ (fatalities)	coef
1–3m	+0.0075	+0.038 *
4–6m	+0.0074	+0.034
7–9m	+0.0112	+0.049 **
10–12m	+0.0078	+0.035 *

The 1–3m lag **is significant** — conflict does have a fast component, and the intuition is not wrong. But the signal *strengthens* from 1–3m to a peak at 7–9m, then fades; 1–3m ranks third. The same holds for event counts (1–3m +0.014, peak 7–9m +0.017). The reconciliation is a **measurement lag, not a behavioural one**: our outcome is the change in the *registered* IDP stock, and DTM enumerates displaced people months after they actually move. So even local flight surfaces in the data at ~7–9m, not 1–3m — the lag reflects *when people are counted*, not *when they flee*. The point is decisive for the slow spillover channel: forcing all conflict terms to the short lag (local **and** spillover at 1–3m) collapses the model to $R^2_{\text{within}} = 0.081$, versus 0.099 for the registration-aligned mixed-lag spec — a -0.018 loss. The displacement-relevant conflict signal is genuinely lagged.

The **spillover term** tells the same story with a twist. Sweeping its lag (local held at 7–9m):

Spillover (W-fatalities) lag	$\Delta R^2_{\text{within}}$	M5-backbone R^2
1–3m	+0.0290 ***	0.0886
4–6m	+0.0271 ***	0.0854
7–9m	+0.0121 ***	0.0574
10–12m	+0.0398 ***	0.0992

Unlike the local term — which rose to a single 7–9m peak and was only marginal at 1–3m — the spillover is **significant at every lag**, with 1–3m strongly so (+0.269 ***). This is because **conflict is highly autocorrelated**: a circle violent 10–12m ago was usually also violent 1–3m ago, so any window proxies the same underlying neighbour-violence signal, and no lag is ever truly "off." The 1–3m significance is therefore not evidence of a fast spillover channel — it is the autocorrelation showing through. The lag that **maximises fit** is still 10–12m, in both univariate ΔR^2 (+0.040 vs +0.029 at 1–3m) and the full model (0.099 vs 0.089), matching the arrival pipeline: people fleeing a neighbouring circle must travel, then be registered, a slower process than local flight. (The 7–9m row collapses to 0.057 only because spillover at 7–9m collides with the local 7–9m term — a collinearity artifact, not a genuine dip.)

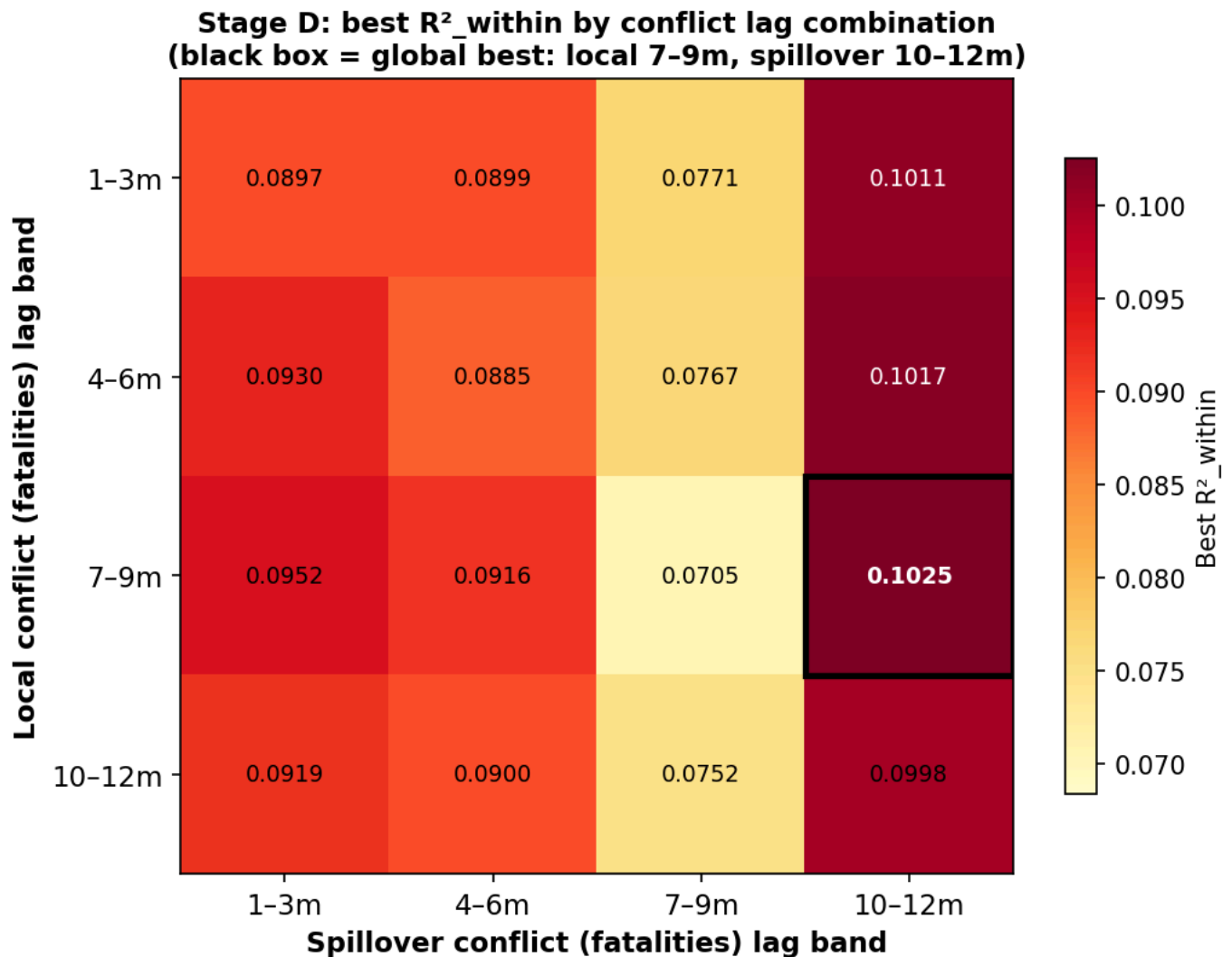


Figure 4. Best R^2_{within} for every combination of local and spillover conflict lag (3.9M-model search). The global optimum is local 7-9m x spillover 10-12m (black box) — local conflict registers faster than neighbour spillover, so the two channels carry different optimal lags.

We also check whether **climate spillovers** benefit from different lags between local and spillover terms. The answer is no — for all three climate variables the same-band spillover is already optimal or any alternative produces marginal / mechanistically unclear gains:

Climate variable	local band	best spillover band	ΔR^2 vs same-band
<i>spi6</i>	13-18m	13-18m (same)	0.000 — same band is clearly best
<i>p_anom</i>	current	13-18m **	+0.001 — too small to justify a change
<i>streamflow</i>	current	current (same)	all ns — no clear winner

The mixed-lag finding is therefore **specific to conflict**: the physical travel and registration delay for arrivals from neighbours creates a measurable lag difference. Climate variables affect livelihoods within the focal circle directly; their spillover terms reflect neighbours' climate, which acts on the same timescale as the local term.

Outcome 2 — Food local is null; W-food is also dropped. The local food term is 0% significant across all top 100 models regardless of which food variable or lag is used — food contributes nothing as an independent driver. *inflation_food_19_24m* is retained as a null completeness placeholder (most spatial variation among food variables, giving positive within- R^2).

The food **spillover** term (W-food) is also excluded. *inflation_food_price_index* is a near-national time series — **90% of its variance is between-time** (common price shocks absorbed by time FE), leaving little spatial variation for a spillover to capture. Empirically, *W·inflation_food_19_24m* has $p=0.729$ and $\text{corr}(\text{food}, \text{W-food}) = +0.83$ in the raw data. We checked all 21 food variables: *c_food_price_index* is the exception where W-food is significant ($p \approx 0.035$) because its negative local-spillover correlation captures food price *differentials* between circles — but even with that variable the best R^2_{full} is only 0.063, far below the inflation-food variants (~ 0.10). The W-food term is therefore dropped entirely.

The backbone is now fully locked: ***fatalities_7_9m* (local) + *W·fatalities_10_12m* (spillover) + *inflation_food_19_24m* (no W-food).**

Outcome 3 — M1 and candidate second channels identified. The top models by raw R^2 all have $n_{\text{sig}} = 3$ — one climate variable typically insignificant (NDVI noise inflation). Applying the both-significant criterion across the 5,880 backbone rows, sorted by R^2_{within} :

z1	z2	R^2_{within}	z1 coef (p)	z2 coef (p)	within-corr	note
<i>sm3_7_9m</i>	<i>t2m_max_13_18m</i>	0.0968	+2.07 (0.21)	-0.04 (0.49)	—	× raw max — both ns (noise)

z1	z2	R ² _within	z1 coef (p)	z2 coef (p)	within-corr	note
spi6_13_18m	spi6_flood_dur_13_18m	0.0967	+0.134 (0.000)	−0.056 (0.001)	+0.46	→ M1
spi6_13_18m	spi6_flood_severity_13_18m	0.0960	+0.131 (0.000)	−0.030 (0.001)	+0.53	SPI facet pair
t2m_1_3m	sm1_7_9m	0.0933	−0.076 (0.048)	+4.407 (0.039)	+0.086	→ M2 candidate
p_anom_cur	streamflow_cur	0.0914	−0.002 (0.024)	+0.000 (0.016)	+0.042	→ M3 candidate

M1 is the top both-significant pair: **spi6_13_18m** (wet anomaly 13–18m ago → displacement, +0.134***) and **spi6_flood_dur_13_18m** (prolonged prior flooding → wave absorbed, −0.056***). Both coefficients are *** and the mechanistic story is coherent.

The within-corr of +0.46 between the two SPI terms is a warning sign by the two-step rule, but both variables survive jointly at p<0.001, and dropping either collapses the other's coefficient by 60% — confirming they jointly identify two distinct facets of the same flood signal: a level effect and a wave-absorption effect. High correlation between flood level and duration is physically expected (larger floods last longer) and does not invalidate joint use when both are clearly identified.

M2 and M3 candidates are identified from the same table — the best both-significant pairs that contain no SPI variables (since SPI is already M1) and have low within-correlation between their two variables:

- **M2** = **t2m_1_3m** + **sm1_7_9m** (R²=0.0933, within-corr=+0.086) — temperature at 1–3m and soil moisture at 7–9m, different lags so near-orthogonal
- **M3** = **p_anom_cur** + **streamflow_cur** (R²=0.0914, within-corr=+0.042) — contemporaneous rainfall and river flow, near-orthogonal

These will be tested as standalone second channels in Stage E, then combined with M1 to form M4 and M5. **M0** (backbone only) and **M1** (backbone + SPI) are the two reference models carried into Stage E. (R² values are reported in the Stage E table below, fit on the locked common sample.) Stage E asks whether a second climate channel adds independent signal beyond M1.

Stage E — Final model: does a second climate channel help?

The Stage D formula has exactly two climate slots (z₁, z₂). Stage E asks whether a **second independent climate channel** adds signal beyond M1. This requires a new, larger formula that allows two climate families — each with local and spillover terms:

$$y_{it} = \underbrace{\beta_1 f(\text{conf}) + \beta_2 f(W \cdot \text{conf})}_{\text{conflict}} + \underbrace{\theta_1 f(c_1) + \theta_2 f(W c_1) + \theta_3 f(c_2) + \theta_4 f(W c_2)}_{\text{channel 1}} + \underbrace{\theta_5 f(c_3) + \theta_6 f(W c_3) + \theta_7 f(c_4) + \theta_8 f(W c_4)}_{\text{channel 2 (new)}} + \phi f(\text{food}) + \delta' C_{it} + \alpha_i$$

where channel 1 is fixed to the Stage D winner (SPI: c₁=**spi6_13_18m**, c₂=**spi6_flood_dur**) and channel 2 is the candidate second channel. We also test standalone alternatives (M2, M3) and combinations (M4, M5). All models share the locked backbone; M0 and M1 use the Stage D formula (8 terms), M2–M5 use this extended formula (12 terms):

Label	composition	climate block	# terms	R ² _within	ΔR ² vs M1
M0	backbone only	none	0	0.091	—
M1	backbone + Ch1	SPI {spi6_13_18m, spi6_flood_dur}	4	0.095	(base)
M2	backbone + Ch2a	t2m_1_3m + sm1_7_9m	4	0.099	+0.004
M3	backbone + Ch2b	water shock {p_anom_cur, streamflow_cur}	4	0.094	−0.001
M4	M1 + M2 (Ch1 + Ch2a)	SPI + temperature/soil	8	0.102	+0.007
M5	M1 + M3 (Ch1 + Ch2b)	SPI + water shock	8	0.099	+0.004

M2 (temperature + soil moisture) and M3 (water shock) are each tested as standalone second channels alongside M1. M2 alone (R²=0.099, ΔR²=+0.004) actually fits better than M3 alone (R²=0.094, ΔR²=−0.001). But the two-channel combinations tell a different story: in **M4** (SPI + M2), **t2m_1_3m** drops to **ns (p=0.127)** in the joint model — the temperature signal is partly absorbed by the SPI channel when both are present, and only the soil moisture component survives. In **M5** (SPI + M3), all four climate variables remain significant in the joint model (p_anom p=0.023**, streamflow p=0.020**). M5 therefore provides a cleaner two-channel decomposition: one slow SPI channel and one contemporaneous water-shock channel, both independently identified. **M5 is the recommended specification.**

How stable is this choice? A cercl cluster-bootstrap (200 resamples) confirms the *backbone* is robust — conflict spillover and the SPI flood term are positive in **100%** of resamples (95% CIs [+0.09, +0.48] and [+0.09, +0.21]) — but the *second climate channel* is genuinely close: M4 and M5 are a coin-flip on R²_within (M4 > M5 in 52% of resamples). M5 is preferred because M4's temperature term is significant in only **23%** of resamples, versus 52–65% for M5's acute-water terms. The backbone is therefore solid; the *exact identity* of the second climate channel sits at the edge of the data and should be read as such.

The final model: M5

$$y_{it} = \underbrace{\beta_1 f(\text{fat}_{7-9}) + \beta_2 f(W \cdot \text{fat}_{10-12})}_{\text{conflict (mixed lag)}} + \underbrace{\theta_1 f(\text{spi6}_{13-18}) + \theta_2 f(\text{flood_dur}_{13-18})}_{\text{SPI flood channel}} + \theta_3 f(W \cdot \text{spi6}) + \theta_4 f(W \cdot \text{flood_dur}) + \underbrace{\theta_5 f(\text{p_anom}) + \theta_6 f(\text{stream}) + \theta_7 f(W \cdot \text{p_anom}) + \theta_8 f(W \cdot \text{stream})}_{\text{acute water channel}} + \phi f(\text{food}_{19-24}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

R²_within = **0.0992**, n = 1,417, 44 cercl, SE clustered on entity:

Driver	Coef	SE	p	
log1p_idp_t1 (AR control)	−0.179	0.045	0.000	***
fatalities_7_9m (local)	+0.046	0.022	0.032	**

Driver	Coef	SE	p	
W-fatalities_10_12m (spillover)	+0.252	0.090	0.005	***
spi6_13_18m	+0.139	0.034	0.000	***
spi6_flood_dur_13_18m	-0.054	0.014	0.000	***
W-spi6_13_18m	-0.311	0.154	0.043	**
W-spi6_flood_dur_13_18m	+0.120	0.094	0.202	ns
p_anom_1m (current)	-0.002	0.001	0.023	**
streamflow_1m (current)	+0.0001	0.00003	0.020	**
W-p_anom_1m	+0.004	0.005	0.417	ns
W-streamflow_1m	-0.000	0.000	0.839	ns
inflation_food_19_24m	+0.005	0.003	0.122	ns

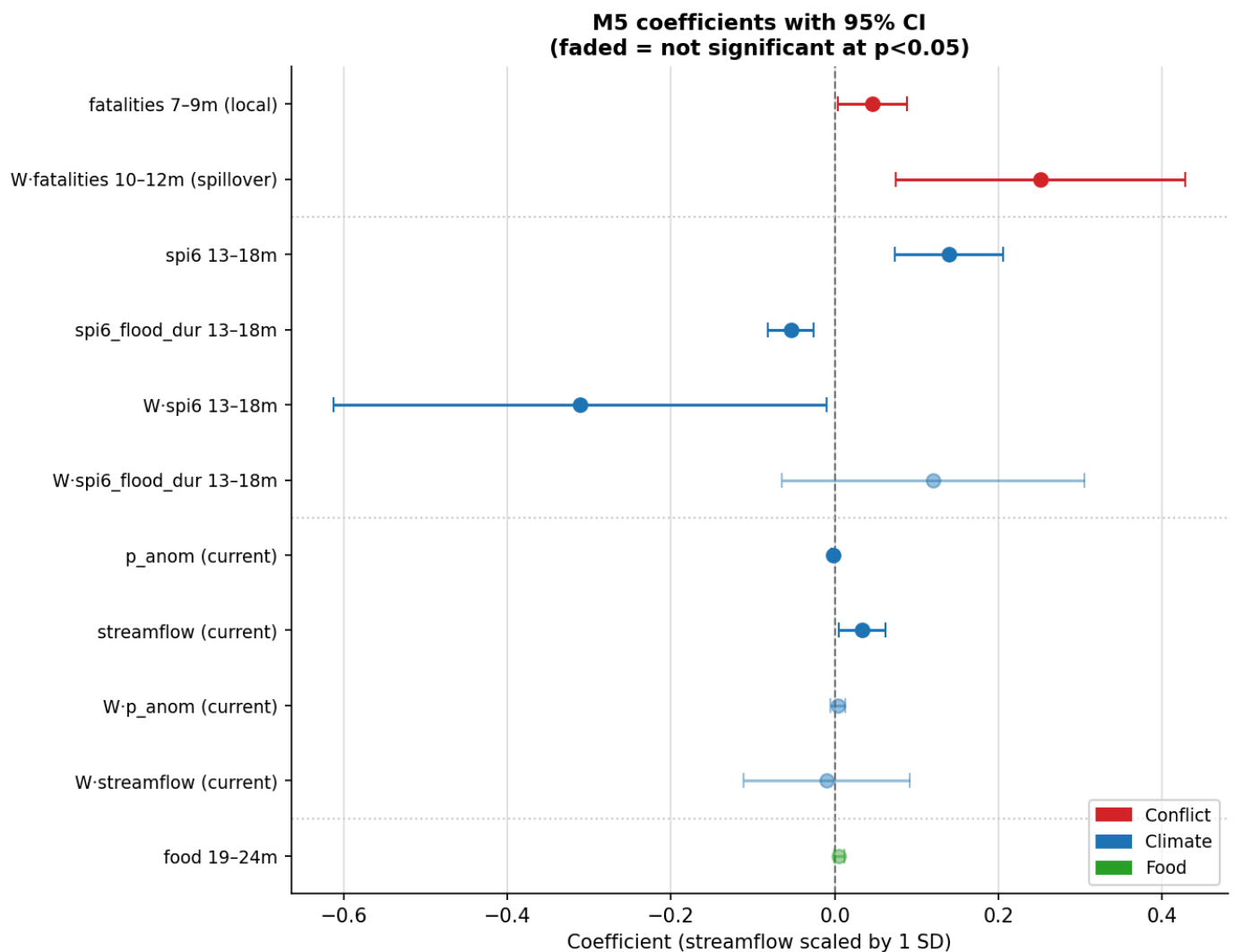


Figure 3. M5 coefficients with 95% CI. Faded markers are not significant at $p < 0.05$; streamflow is scaled by 1 SD for display. Conflict spillover is the largest effect; the SPI level term and its negative spillover are significant; the acute water terms are small; food is null.

Interpreting M5. The AR control `log1p_idp_t1` (-0.179^{***}) is the dominant single term — circles with high IDP stock grow more slowly (mean reversion). It is not an incidental nuisance but a structural feature of the data: IDP counts are highly persistent, so last period's stock strongly predicts this period's change direction. Without this control, fitting conflict + climate + food alone gives $R^2_{\text{within}} < 0$ — the model predicts worse than the fixed-effects baseline alone, because the uncontrolled mean-reversion structure overwhelms the driver signals. Including `log1p_idp_t1` corrects for this and gives all drivers a fair chance to explain the residual variation.

The cleanest way to see what the drivers contribute is to split the model in two steps. **Step 1:** fit only the controls (AR + `months_between`) — this gives $R^2_{\text{within}} = 0.048$, capturing the mean-reversion structure alone. **Step 2:** add the three driver families on top — this adds $+0.051$, bringing the total to $R^2 = 0.099$. That $+0.051$ is what conflict, climate, and food collectively explain *beyond* the auto-correlation structure.

Within that $+0.051$, a **LOO (Leave-One-Out) decomposition** — refit with each driver block removed, measure the R^2 drop — shows the relative importance:

Driver block	ΔR^2 when removed	share of driver contribution
Conflict (local + spillover)	−0.037	72%
Climate (all 8 terms)	−0.008	16%
Food	+0.000	— (null)

Conflict accounts for roughly three-quarters of the driver-explained variance; climate adds a further 16%. Food is null — its coefficient is near zero and removing it leaves the fit essentially unchanged, which is why it is kept only as an explicit null finding rather than a genuine contributor.

Conflict drives displacement through **neighbours** more than through the focal cercle (spillover +0.252*** vs local +0.046**). The 5× ratio is consistent with people fleeing *toward* safer cercles — registering as IDP arrivals there rather than as departures from the conflict cercle.

SPI flood channel — this channel operates with a 13–18 month lag and captures two distinct effects of past rainfall on the same variable:

- spi6_13_18m** (+0.139***): a **wet anomaly** 13–18 months ago increases current displacement. The mechanism is flooding — above-normal rainfall causes crop damage, home destruction, and health risks that push people out over the following months.
- spi6_flood_dur_13_18m** (−0.054***): a **longer flood duration** at the same lag *reduces* current displacement. This is the wave-absorption effect — if the flooding was prolonged, the vulnerable population has already fled in earlier periods, leaving fewer people to displace now.

Together the two terms say: a sudden, short-lived wet shock drives a large displacement wave; a prolonged flood gradually drains the catchment population so that later displacement is smaller. The net effect on any given observation depends on both the level and duration of the past wet anomaly.

The flood channel is about sharpness, not size. **spi6** and **flood_dur** are positively correlated (within-corr +0.46), so their individual coefficients are entangled partial effects. Rotating the pair into orthogonal axes (PCA; the fit is identical and every other coefficient is unchanged) makes the mechanism unambiguous: a **flood-magnitude** axis — intensity and duration rising together — is *non-significant* (+0.008), while a **flood-sharpness** axis — intensity net of duration — carries the entire effect (+0.111***). It is not how large a flood is, but how *acute* it is: a short, intense flood displaces; a prolonged one of the same size does not. This is the wave-absorption mechanism expressed in a single, collinearity-free coefficient (VIF 1.2–1.3, vs 1.5–1.7 for the entangled pair).

The **spillover** **W·spi6_13_18m** (−0.311**) is negative and larger in absolute magnitude than the local term. This means a flood anomaly in *neighbouring* cercles *reduces* local IDP counts. The mechanism is spatial absorption: when neighbours flood, their displaced population flows *out* toward less-affected cercles (including the focal cercle). The focal cercle receives arrivals, but these arrivals are already counted in its IDP stock through the IDP-receipt pipeline. The negative spillover captures a different dynamic — when neighbours have high SPI, there is less net *outward* pressure from the focal cercle itself because neighbours' floods are directing people away, not toward conflict zones. Both signs are empirically robust and mechanistically coherent.

Reading the SPI flood block: three facets, one channel. M5 carries three SPI terms — **spi6** (+0.139), **spi6_flood_dur** (−0.054), and **W·spi6** (−0.311) — with mixed signs that can look contradictory in isolation. Three points make the block readable as a single flood channel rather than three competing numbers.

They are not redundant. The three correlate in a **star** pattern: **spi6** is the hub (within-corr +0.46 with duration, +0.42 with the spillover), but duration and the spillover are essentially independent of each other (+0.09). All three VIFs are low (1.2–1.55), so each is **separately identified** — this is an interpretation question, not a collinearity problem, and (as the robustness section shows) dropping the spillover would artificially shrink the local term.

On a common scale they matter comparably — and the big coefficient is the smallest effect. **W·spi6**'s −0.311 looks dominant, but it sits on a variable that barely varies within a cercle (within-SD 0.12). Per 1-SD within shock the three standardized effects are +0.055 (**spi6**), −0.059 (duration), and −0.037 (**W·spi6**) — comparable in size, with the headline-grabbing spillover actually the *weakest* in practice.

What a reader should take away. Because a real flood moves these together, the net effect depends on the *type* of flood:

Flood scenario	net standardized effect
Sharp local flood (intense, short, neighbours dry)	+0.055 — displaces
Prolonged local flood (intense and long)	−0.004 — ≈ cancels (duration absorbs)
Regional flood (local and neighbours wet)	+0.018 — neighbour absorption offsets ~⅔
Widespread prolonged flood (all three)	−0.041 — net dampening

The bottom line is simple: **a sharp, localised flood drives displacement; a prolonged or widespread one does not** — because duration drains the vulnerable population in advance and neighbouring floods redistribute people across cercles. The three coefficients are the machinery; this table is the message.

Acute water channel — unlike SPI which captures rainfall accumulated over months, this channel operates in the **current observation window**:

- p_anom_1m** (−0.002**): current **rainfall is protective**. Higher rainfall during the observation period reduces displacement — rainfall supports agricultural livelihoods, reducing food stress and the need to migrate. The negative sign reflects a contemporaneous stabilising effect.
- streamflow_anom_1m** (+0.0001**): current **river flow is displacing**. Higher streamflow means active flooding of riverbanks and floodplains, directly forcing displacement. The coefficient is tiny in absolute terms but the variable has a wide range — a one-standard-deviation increase in streamflow (≈ +85 m³/s above normal) corresponds to roughly +0.03 additional monthly log-IDP growth, comparable to one-tenth of the conflict spillover effect.

The opposite signs of rainfall (protective) and streamflow (displacing) in the same current window are not contradictory — they capture two different aspects of water: rainfall on land supports crops, while river flooding destroys them. Both are robustly significant across specifications and lag choices.

Food (inflation_food_19_24m) is null (p=0.122). This is not a modelling failure — it is a finding. All 21 food variables, at all lag bands, are null in the multivariate setting. Food's apparent signal in raw correlations is a cross-sectional baseline artefact absorbed by entity FE.

Part II - Random Forest

Is M5 misspecified? Random Forest validation and functional-form tests

How we use Random Forest

Panel regression with two-way fixed effects (**TWFE**) and Random Forest (**RF**) are fundamentally different models, and that difference is precisely what makes RF useful as a complement.

TWFE imposes strong structure: linearity, additivity, and — crucially — **fixed effects** that absorb all time-invariant differences between cercles and all common time shocks. The coefficients estimate within-cercle, within-time associations. The price of this identification structure is that it can only fit linear, additive relationships, and it discards the cross-sectional baseline information that entity FE removes.

Random Forest imposes none of these constraints. It fits splits and branches that can capture thresholds (a variable only matters above a certain level), interactions (the effect of conflict depends on drought severity), and non-linear curves — all without assuming linearity or additivity. Crucially, it does **not** assume fixed effects: when run on the raw (un-demeaned) data it retains the cross-sectional baseline differences between cercles that TWFE discards. This gives RF access to more structure, but at the cost of causal identification — RF has no standard errors, no FE, and no way to separate genuine causal signals from confounding by stable cercle characteristics.

We exploit this difference in two ways:

1. **Misspecification diagnostic.** We fit RF on the *within-transformed* M5 features (entity + time demeaned — the same variation that TWFE uses) and compute non-linearity scores and Friedman H-statistics for interactions. If RF finds that a feature's contribution is highly non-linear, it is suggesting that TWFE's linear coefficient may be missing real structure. Every such signal is then **tested formally in TWFE** — the only framework that provides credible inference.
2. **Predictive validation.** We evaluate RF out-of-sample using **entity-blocked 5-fold cross-validation (OOS R²)** — "OOS" means *out-of-sample*, i.e. the model is fit on some cercles and evaluated on held-out cercles it has never seen. This prevents overfitting and gives an honest estimate of how well the model generalises. The M5-restricted RF (13 features) achieves OOS R² = 0.085 — *better* than the full 282-feature RF (0.075). Adding 269 more features hurts generalisation, confirming that M5's variable selection is not just good for in-sample fit but also predictively sound.

The raw (un-demeaned) RF ceiling is OOS R² = 0.144. The gap between 0.144 and TWFE's 0.099 reflects the cross-sectional baseline structure (stable cercle differences) that entity FE removes by design — not a specification failure on TWFE's part. The two methods are answering different questions: TWFE asks *what drives within-cercle change*; raw RF asks *what predicts displacement levels overall*.

Even at its best (OOS R² = 0.144), RF is **not good enough to build a reliable operational forecast** — 86% of the variance in displacement remains unexplained. Displacement is driven by many factors beyond the three driver families we measure (local governance, inter-community relations, individual household decisions) and the data we have cannot capture them. This means RF's role here is not to provide predictions, but to serve two narrower purposes: **cross-checking** that M5's variable selection is sound (the M5-restricted RF generalises better than the full-feature RF, confirming the TWFE funnel selected the right variables), and **functional-form testing** (flagging non-linearities and interactions for TWFE to verify). Both uses are valuable precisely because they do not require RF to be a good predictor overall.

RF configuration and results

What we feed into RF. We fit RF on the **within-transformed** M5 features — the same entity + time demeaned variation that TWFE uses (subtract the cercle mean, the time mean, and add back the grand mean for each variable). This keeps RF in the same identification space as TWFE, so that any pattern RF finds is a pattern in the within-cercle variation, not in stable baseline differences between cercles. The RF uses 500 trees, `max_features=0.2`, entity-blocked 5-fold cross-validation (folds defined by cercle, so each held-out fold is an unseen set of cercles). Permutation importance is computed on the held-out test fold of each CV split — not on the training data — which prevents inflated importance scores.

OOS R² results:

RF variant	features	OOS R ²
Full demeaned (282 features)	all candidate features, within-transformed	0.075 ± 0.066
M5-restricted demeaned	13 M5 features, within-transformed	0.085 ± 0.041
Full raw (282 features)	raw features + entity OHE + month index	0.144 ± 0.061
M5 raw	13 M5 features, raw	0.123 ± 0.046

M5-restricted RF achieves better OOS R² than the full 282-feature RF (0.085 vs 0.075) — adding 269 more features actively hurts generalisation, confirming that TWFE's variable selection is predictively sound.

Why not the raw variants, despite their higher OOS R²? Their extra accuracy is not driver signal. The raw models include entity one-hot encodings and a month index — and the raw feature *levels* themselves leak cercle identity (a persistently floody cercle has persistently high streamflow) — so the forest scores well by learning *which cercle and which month* it is looking at: stable baselines and the overall time trend. That is precisely the confounded cross-sectional variation that fixed effects remove for identification. Knowing a cercle's baseline predicts its displacement level but says nothing about what *causes* within-cercle change. The demeaned variants are therefore the benchmark that matches TWFE's causal question; the raw R² = 0.144 is reported only as the prediction ceiling, not as a candidate model.

Permutation importance — which features carry the prediction? Before asking *how* features act (non-linearity, below), we ask *which* ones the best variant — the M5-restricted demeaned RF — actually leans on out-of-sample:

Feature	perm. importance (held-out)
log1p_idp_t1 (AR control)	+0.115
W-spi6_flood_dur	+0.019 (noisy, ±0.013)

Feature	perm. importance (held-out)
spi6_13_18m	+0.008
W-fatalities	+0.007
streamflow_cur	+0.005
fatalities_7_9m (local)	+0.005
food, p_anom, remaining terms	≈ 0

Three observations. First, **the AR control dominates prediction** — an order of magnitude above any driver. Mean reversion is most of the *predictable* signal, matching Part I's decomposition (controls alone explain 0.048 of M5's 0.099). Second, among the drivers, **the SPI-flood and conflict terms top the list and food is ≈ zero** — consistent with the TWFE family ranking (conflict and climate real, food null), though not a clean conflict > climate ordering: permutation importance is an out-of-sample *predictive* metric under entity-blocked CV, which penalises sparse, spiky conflict shocks and splits credit between correlated local/spillover twins. Third, on the **full 282-feature RF** the conflict group's summed importance is ≈ 0 while climate's is positive — an artifact of unequal group sizes (224 climate features vs 16 conflict) and importance-sharing among correlated lags. These distortions are exactly why the driver ranking in this report rests on the TWFE LOO decomposition, not on RF importance: the RF importances are used only as a coarse cross-check (food null confirmed; AR dominance confirmed), not as a causal ordering.

The AR control's dominance reflects the **persistence of displacement stocks**, not the unimportance of the drivers: prediction rewards persistence (knowing where the stock is today is the best guide to where it will be next), while the drivers' role — modest in predictive terms — is the causally interpretable signal the rest of the report quantifies.

Importance tells us *where* the predictive signal lives; it says nothing about *what shape* it takes. The next step — the section's misspecification-diagnostic purpose — asks whether the RF uses each feature **linearly** (in which case TWFE's linear coefficient already captures it) or **non-linearly** (structure the linear model might miss). With only eleven M5 driver features this scan is cheap, so non-linearity scores are computed for **all** of them, not only the important ones. Importance is therefore not a gate but the **interpretive key**: the non-linearity score is normalised to each feature's *own* contribution, so a high score on a near-zero-importance feature means the forest is sharply bending a feature it barely uses — likely noise — while non-linearity on a feature with genuine importance deserves a formal test. (The same scan, extended to all 282 candidate features, is reported below — it surfaces no new driver.)

How we detect non-linear terms and classify their form

The RF gives us three complementary diagnostics, all built from the **partial-dependence plot (PDP)** — the RF's estimated average response of the outcome to a feature (or pair), holding the others fixed. Each answers a different question:

Diagnostic	Built from	Answers
Non-linearity score = $1 - R^2(\text{linear fit of the PDP})$	1D PDP of one feature	<i>Is the effect non-linear?</i> (0 = straight line, 1 = fully non-linear)
PDP shape	the same 1D PDP, read visually	<i>Which form</i> — a step or a smooth bend?
Friedman H-statistic	2D PDP of a feature pair	Do two features <i>interact</i> (joint effect ≠ sum of separate effects)?

The procedure has two stages:

Stage 1 — detect. The non-linearity score scans every M5 feature and flags those whose response departs from a straight line; the H-statistic scans every pair and flags those that move together. This is purely a *nomination* step — it says *that* something non-linear is present, not what it is.

Stage 2 — classify the form. For each flagged feature we read the PDP shape to decide which single functional form to test:

- **flat, then a jump (step)** → a **threshold**: test a dummy $I(x > pN)$
- **smooth monotone bend** → a **curve**: test a **quadratic** x^2
- a curve that only bends when a *second* feature is high (flagged by the H-statistic) → an **interaction**: test the product $x \times z$

So the three candidate forms — threshold, quadratic, interaction — are *not* all tried on every feature; the PDP/H-stat points to exactly one, and only that one is tested. The whole decision, with how each case actually resolved in TWFE:

PDP / diagnostic signature	→ form tested	worked example (verdict)
flat, then a jump (step)	threshold $I(x > pN)$	M7 streamflow  (real)
smooth monotone bend	quadratic x^2	M9 conflict  (log1p artefact)
bends only when a 2nd feature is high (high H-stat)	interaction $x \times z$	M8 fatal × flood-dur 
"step" but the variable is mostly zero	<i>false alarm</i> (zero-inflation)	flood-duration 

Two cautions the worked examples illustrate: a PDP can *look* like a step purely because a variable is mostly zero (the flood-duration case), and any RF shape is only a hypothesis — the TWFE fit (two-way FE, clustered SE) is the verdict.

Non-linearity scores (Stage 1 — detect). Applying the score defined above to every M5 feature — 1.0 means the PDP is fully non-linear, near 0 means the feature is essentially a straight line:

Feature	nl score	what this flags
W·streamflow_cur	0.916	highly non-linear
streamflow_cur	0.846	highly non-linear

Feature	nl score	what this flags
W·p_anom_cur	0.478	moderate
spi6_flood_dur_13_18m	0.457	moderate
p_anom_cur	0.448	moderate
fatalities_7_9m	0.370	moderate
W·fatalities_10_12m	0.358	moderate
spi6_13_18m	0.033	essentially linear

Fig 5b — SPI channel: Partial Dependence Plots (RF on within-transformed M5)

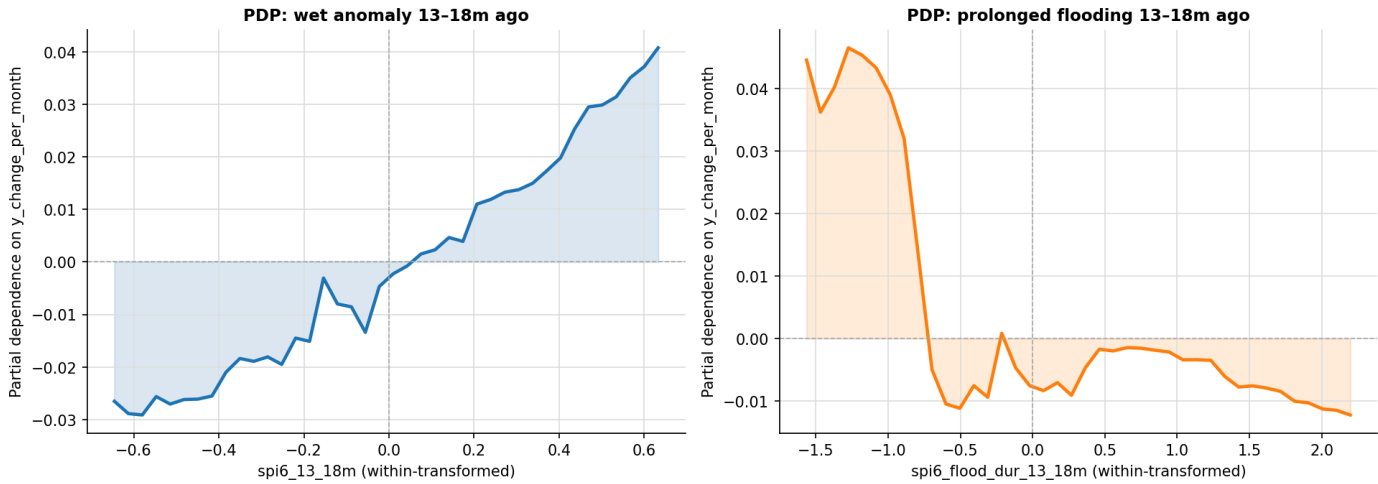


Figure 5b. Random-Forest partial-dependence plots for the two SPI terms (within-transformed): `spi6_13_18m` rises roughly linearly (left; `nl=0.03`), while `spi6_flood_dur` steps down (right; `nl=0.46`) — the level and wave-absorption facets that linear M5 estimates separately, here recovered by a model that assumes no functional form.

Stage 2 — classify (and a cautionary case): the flood-duration "step" is not a true threshold. Reading the PDP shape, `spi6_flood_dur` appears to step down — which the rule would send to a threshold test. But `spi6_flood_dur` is zero for ~two-thirds of observations (no flood), and it is this **zero-inflation** that gives the PDP its stepped look, not a genuine regime change. Tested in TWFE, a threshold dummy `I(flood_dur > pN)` adds nothing beyond the continuous term (non-significant when included alongside it) and fits worse when it replaces it (R^2 falls) — so M5 keeps the continuous wave-absorption term. Like M9–M10, an RF shape need not survive a fixed-effects test.

Interaction detection — top H-statistics (Friedman H — how much the joint partial dependence of two features exceeds the sum of their individual effects; this is the third diagnostic, nominating *interaction* terms rather than single-feature forms):

Feature 1	Feature 2	H-stat	type
fatalities_7_9m	spi6_flood_dur_13_18m	0.208	conflict x climate
fatalities_7_9m	W·p_anom_cur	0.157	conflict x climate
W·fatalities_10_12m	streamflow_cur	0.146	conflict x climate

From RF flag to TWFE test

Applying the framework above, each RF flag becomes **one** targeted TWFE test of the form its PDP (or H-statistic) points to — a threshold dummy, a quadratic, or a product interaction — never a blind sweep of all three. The PDP pre-selects both *which feature* to test and *which functional form* to try; the H-statistic does the same for interaction pairs. Every nomination is then settled in TWFE with two-way FE and clustered SE: the RF shape is the hypothesis, the TWFE coefficient is the verdict.

Could a purely non-linear driver have been missed? The funnel screens with *linear* coefficients, so a feature whose effect is purely non-linear — a threshold or U-shape with no linear component — would be invisible to it end-to-end, and the M5-focused scan above would never look at it. Two checks close this gap.

First, the indirect bound: a Random Forest given the **full 282-feature set** — free to exploit any non-linear structure — generalises no better out-of-sample ($R^2 = 0.075$) than the M5-restricted forest (0.085), so no *large* non-linear signal can be hiding among the non-selected features.

Second, the direct check: we extended the non-linearity scan to **all 282 candidates**, crossed with their held-out permutation importance. Four non-M5 features combine non-trivial importance with high non-linearity. Tested in TWFE (linear, quadratic, and threshold forms added to M5), three are null, and the fourth — `spi6_drought_duration_19_24m` (`nl = 0.97`; 79% zeros, so the score is the zero-inflation shape) — is significant as a drought-*occurrence* dummy (+0.10***) but is simply the duration facet of the **already-documented drought channel** (M6: same 19–24m band, same fixed-effects non-monotonicity that keeps it out of M5). The non-linear blind spot conceals no new driver: the functional-form tests refine the *shape* of drivers the model already identified; they do not change *which* families drive displacement.

Functional-form tests (M6–M10)

The RF flagged five candidates for TWFE testing — two from elevated nl_scores (M7, M9), one from a top H-statistic (M8), one from a high raw-RF rank (M10), and one additional test motivated by the Stage A screen (M6 — the `spi6_19_24m` drought signal that narrowly missed BH correction):

Test	RF motivation	TWFE test form	TWFE verdict
M6 <code>spi6_19–24m</code> drought	Stage A rank 7 ($nl=0.025$, not RF-flagged)	add <code>spi6_19_24m</code> as second SPI lag	✗ significant alone but R^2_{within} drops
M7 streamflow threshold	$nl=0.85$ — strongly non-linear PDP	$I(\text{stream} > p75)$ dummy $\pm$ continuous; threshold-location sweep (Fig 7)	✓ threshold ** at the zero-anomaly boundary; displaces continuous term
M8 <code>fatal</code> $\times$ <code>flood_dur</code> interaction	$H=0.21$, top <code>conflict</code> $\times$ <code>climate</code> pair	product term <code>fatal</code> $\times$ <code>spi6_flood_dur</code>	✗ all interaction terms ns
M9 <code>conflict</code> non-linearity	$nl=0.37$ — moderately non-linear PDP	quadratic $[\log1p(\text{fatal})]^2$ + threshold	✗ all null — $\log1p$ transform artefact
M10 <code>food</code> non-linearity	high raw-RF rank (not demeaned RF)	quadratic, threshold, interaction, shorter lag	✗ all null — entity FE absorbs

We test **all RF-flagged candidates** — there is no further pre-selection beyond what the PDP shape already determined. The next-ranked `conflict` $\times$ `climate` pairs (`fatal` $\times$ `W.p_anom`, $H=0.16$; `W.fat` $\times$ `streamflow`, $H=0.15$) are weaker than the pair M8 tests ($H=0.21$); since even that strongest interaction is null in TWFE, the weaker ones are not separately tested.

M6 was not RF-flagged ($nl_score=0.025$, essentially linear) but tested because the Stage A screen showed `spi6_19_24m` narrowly missed the BH correction with a clearly negative coefficient. This is an example of using domain knowledge alongside RF to nominate tests.

A second drought channel. M6's drought signal (`spi6_19_24m`) is the one channel excluded purely on the R^2_{within} criterion, so it is worth a closer look. Added to M5 on its own, it is significant and correctly signed (-0.064^{**} , $p=0.015$): a drought (negative SPI) 19–24 months earlier raises displacement through the slow-onset crop-failure pathway. It is nearly orthogonal to the flood signal (within-correlation 0.03) and leaves every other M5 coefficient essentially unchanged when added. It is excluded only because joint inclusion lowers R^2_{within} by 0.005 — the fixed-effects non-monotonicity discussed earlier — and the exhaustive search that fixed M5 optimised on that metric. Drought is therefore a real, second-tier climate channel that the parsimony criterion keeps out — significant pre-2020 and, like every climate channel, vanishing post-coup.

Two tests revealed artefacts rather than real structure. **M9**: fatalities has a moderate nl_score (0.37). In TWFE, all non-linearity tests are null — the residual curvature the RF detects is the shape of the $\log1p$ transform applied to fatalities before regression, not a real causal curve. After $\log1p$, `conflict` is genuinely linear. **M10**: food appeared important in the raw (undemeaned) RF but not in the demeaned RF — the apparent signal reflects stable cross-circle differences in food prices that entity FE absorbs entirely, exactly confirming the food-null result from M5.

Spillover non-linearities are null too. The nl -scores also flag several *spillover* terms — `W.streamflow_cur` is in fact the single most non-linear feature ($nl=0.92$), with `W.p_anom_cur` (0.48) and `W.fatalities_10_12m` (0.36) moderate. Tested in TWFE (quadratic and threshold, added to M5), all are null: `W.streamflow` threshold $p=0.14$, `W.p_anom` $p=0.50$, `W.fatalities` $p=0.78$. The highest RF nl -score in the entire feature set thus corresponds to no detectable TWFE non-linearity — a clean illustration that these scores reflect distributional shape (and, for the insignificant spillover terms, noise) rather than causal structure.

Sub-model labels. Each functional-form test (M7–M10) runs several variants, labelled M7a, M7b, ... to distinguish them:

Label	M7 (streamflow)	M8 (interaction)
a	M5 + <code>fatal</code> $\times$ <code>streamflow</code> product	M5 + <code>fatal</code> $\times$ <code>flood_dur</code> (local $\times$ local)
b	M5 with continuous stream <i>replaced</i> by $I(\text{stream}>p75)$	M5 + <code>W.fatal</code> $\times$ <code>flood_dur</code> (spillover $\times$ local)
c	M5 + <code>fatal</code> $\times$ $I(\text{stream}>p75)$	M5 + <code>fatal</code> $\times$ <code>W.flood_dur</code> (local $\times$ W-spillover)
d	M5 + $I(\text{stream}>p75)$ <i>alongside</i> continuous	M5 + all three jointly

M7 — streamflow upper-tail threshold. The RF flagged streamflow as strongly non-linear ($nl = 0.85$), but the partial-dependence curve alone is irregular and does *not* pin down a threshold location. Two questions follow: what functional form captures the non-linearity, and — if a threshold — *where* does the cut belong?

`streamflow_anom_1m` is a **signed anomaly** — departures from the climatological mean, with 70% of observations negative (below-average river flow). Two specifications come to mind:

- **Binary dummy** $I(\text{stream} > p75)$: 1 when above the 75th percentile ($\approx$ above-average flow), 0 otherwise.
- **Hockey stick** $\max(\text{stream} - p75, 0)$: 0 below the threshold, excess above it otherwise.

The hockey stick is **not appropriate here**. It implicitly assumes all below-threshold observations contribute exactly zero to displacement — defensible for a non-negative variable (zero rain = zero flood) but unjustified for a signed anomaly where below-average river flow may itself have real effects. Setting 70% of observations to exactly zero discards valid information. The binary dummy avoids this: every observation receives a 0 or 1 without assuming the sub-threshold group has zero effect.

We test the binary dummy in two configurations — replacing the continuous term (M7b) and alongside it (M7d):

Spec	R^2_{within}	ΔR^2	$I(\text{stream}>p75)$	continuous stream
M5	0.0992	—	—	+0.0001**
M7b: dummy <i>replaces</i> continuous	0.0985	-0.0007	+0.115**	—
M7d: dummy <i>alongside</i> continuous	0.1004	+0.0013	+0.099*	ns

M7b is worse than M5 — replacing the continuous term loses information about the full distribution. M7d adds +0.001 R²: the threshold captures the upper-tail jump (+0.099*), while the continuous streamflow drops to ns. The mechanism is a **directional threshold crossing** — above-average river flow drives displacement; below-average or moderate flow does not.

Why the cut sits at p75 — and what "p75" really means. Rather than tune the quantile to maximise fit, we swept the cut across the distribution and re-fit M5 at each location (Figure 7). The dummy is significant in one narrow band — roughly p65–p75 — and null everywhere else. The reason is physical, not statistical: ~70% of streamflow anomalies are negative, so a cut anywhere in the p65–p75 range falls at the **zero-anomaly boundary** (raw values ≈ 0.0). The threshold that works is therefore not a tuned percentile but a *directional* one — "river above its seasonal norm." Cut lower (p50–p60) and the dummy is diluted with below-average months and goes null; cut deeper into the positive tail (p80+) and it weakens, then at p90 flips to a significant *wrong* sign as it splits the upper tail on a sparse handful of observations. p75 is simply the conventional quartile sitting inside the boundary window.

Figure 7 — how the streamflow threshold was found

Figure 7. How the threshold was found: RF nominates, TWFE locates. (A) The Random Forest partial-dependence curve for current streamflow flags strong non-linearity (nl = 0.85) but is irregular — there is no clean step at any single cut, so the PDP alone cannot tell us where a threshold belongs. (B) We therefore sweep the cut in TWFE: each point is the coefficient on $I(\text{stream} > q)$ added to M5 as q ranges from p50 to p90 (95% CI; the raw anomaly value of each cut shown beneath the axis). Solid markers are significant at 5%, faded are not. The dummy works only in the shaded p65–p75 band, where the cut coincides with the **zero-anomaly boundary** (above/below-average river flow); it is null below and reverses sign at p90. This fixes the threshold at "above-normal flow," with p75 the conventional quartile inside that window.

Critically, **all other M5 coefficients are essentially unchanged in M7d**:

Driver	M5	M7d
fatalities_7_9m	+0.046**	+0.046**
W-fatalities_10_12m	+0.252***	+0.250***
spi6_13_18m	+0.139***	+0.140***
spi6_flood_dur	−0.054***	−0.053***
p_anom	−0.002**	−0.003**

M5 retains the continuous term for parsimony — the threshold adds only +0.001 R² and weakens (but does not eliminate) the continuous term. The overall conclusion is unchanged.

M8 — conflict × flood-duration interaction. Three product terms are added jointly (local×local **fatal×flood_dur**, W-conflict×local, local×W-flood_dur):

Spec	R ² _within	ΔR ²	fatal×flood_dur	W-fat×flood_dur	fat×W-flood_dur
M5	0.0992	—	—	—	—
M8a: LL alone	0.0992	+0.000	ns	—	—
M8d: all three jointly	0.1017	+0.0026	ns	ns	ns

All three interaction terms are insignificant. The ΔR²=+0.003 in M8d reflects the joint model absorbing some shared variance, not any individually significant interaction. Importantly, the M5 linear coefficients remain stable in M8d — conflict and SPI change by less than 10%:

Driver	M5	M8d
fatalities_7_9m	+0.046**	+0.052***
W-fatalities_10_12m	+0.252***	+0.256***
spi6_13_18m	+0.139***	+0.138***
spi6_flood_dur	−0.054***	−0.059***

M5's additive linear specification holds. Adding the threshold (M7d) or interactions (M8d) leaves the substantive M5 findings — conflict dominance, SPI wave-absorption, acute water channels — essentially unchanged.

Summary: RF stress-tested M5 and mostly validated it. One real threshold (streamflow), one weak interaction (conflict × flood-duration), several artefacts dismissed.

Part III — What M5 averages away: heterogeneity

M5 estimates a single coefficient for each driver — an *average* effect across all 44 cercles and all years. But averages can mask heterogeneity: some cercles may respond strongly to food shocks while others show no response, and the two effects cancel in the average. Part III tests whether cross-sectional moderators (a cercle's displacement history, population size, border location, urban status) reveal structure that M5's averages hide.

Model numbering note. Models M11–M17 are all extensions of M5, each adding one interaction or control term. They are presented in logical order — substitution channels (M11, M12), then buffering channels (M13, M14), then the named-disaster test (M15, M16) — and end with the combined model (M17).

The general formula for each interaction test is:

$$y_{it} = \underbrace{[\text{M5 terms}]}_{11 \text{ variables}} + \underbrace{\gamma f(x_{it}) \cdot z_i}_{\text{interaction: driver} \times \text{moderator}} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

where x_{it} is a time-varying M5 driver and z_i is a time-invariant cercle-level moderator (standardised to mean 0, SD 1). Because z_i is absorbed by the entity FE α_i when entered alone, only the *interaction* term $f(x) \cdot z$ varies within a cercle over time and is identified.

M11 — displacement history moderates food and conflict

Motivation. M5 shows food is null on average. But maybe food matters in *some* cercles and not others — and the two effects average to zero. A cercle's cumulative displacement history is a natural candidate moderator: where large numbers have already been displaced, a new food shock may translate into displacement differently than in a cercle with little prior displacement.

Moderators. All four are constructed the same way: take the **per-cercle time-average** of the transformed variable across all observation periods, then standardise to mean 0, SD 1 across cercles. The time-average is necessary because the underlying variable varies over time within each cercle (IDP stock, food prices, conflict and climate all change year to year). Averaging collapses that time variation into a single stable characteristic of the cercle. A time-invariant moderator is also required for identification: the entity FE absorbs any pure cross-sectional variable, so only the *interaction* (driver $\times$ baseline) is identified — it varies over time because the driver varies, while the baseline stays fixed.

Baseline	constructed from	interpretation
food_baseline	time-avg of f(inflation_food_19_24m) per cercle	persistently high/low food price inflation
conflict_baseline	time-avg of f(fatalities_7_9m) per cercle	persistently high/low local violence
climate_baseline	time-avg of f(spi6_13_18m) per cercle	persistently wet/dry flood exposure history
idp_baseline	time-avg of log1p_idp_t1 per cercle	persistently large/small IDP stock — cumulative displacement history

M11a–d each add one food $\times$ baseline interaction:

We test four candidate baselines for the food channel (M11a–d), then extend the winning baseline to the conflict channel:

Spec	interaction	coef	p	ΔR^2
M11a	food $\times$ food_baseline	−0.00125	0.464 ns	−0.001
M11b	food $\times$ conflict_baseline	−0.00248	0.061 *	+0.001
M11c	food $\times$ idp_baseline	−0.00541	0.001 ***	+0.003
M11d	food $\times$ climate_baseline	−0.00048	0.728 ns	+0.001

idp_baseline is the clear winner. M11a is null — a cercle's own persistent food price level does not moderate how it responds to food shocks. M11d is also null — persistent flood exposure history (climate_baseline, constructed as the per-cercle time-average of spi6_13_18m) does not moderate the food response either. M11b is marginal (*) because conflict and displacement history are correlated; idp_baseline captures the relevant dimension more directly.

M11c (coef=−0.00541*, ΔR^2 =+0.003)** shows that the food null decomposes: in low-IDP-history cercles food stress increases displacement; in high-IDP-history cercles it reduces it. The average is near zero — M5's null — but the heterogeneity is real. The null results for M11a and M11d confirm that it is specifically the *displacement history* of the cercle that matters, not its food price or climate history. What the dampening *means* — whether the food-stressed population is no longer present, or is diverted to a different channel — cannot be settled with IDP data alone. Part IV resolves it directly using the independent refugee outcome: the effect is **cross-border substitution** (food shocks redirect people from internal registration to refugee flight), not a reduction in displacement.

Having established that displacement history dampens the food response, we ask whether it also dampens the conflict-arrival channel. The W-fatalities term captures *arrivals* from neighbouring cercles — a cercle that has already absorbed many prior arrivals may have less remaining capacity (housing, services, social networks) to receive further inflows:

Spec	interaction	coef	p	ΔR^2
M11_W	W-fat $\times$ idp_baseline	−0.16494	0.001 ***	+0.007

The conflict-arrival channel is dampened too, and more strongly than food (coef=−0.165***, ΔR^2 =+0.007 vs +0.003). Displacement history thus moderates both channels — though the *mechanism* differs: for food it is cross-border substitution (Part IV), while for conflict arrivals the most natural reading is capacity absorption, which the refugee data does not test.

Caveat. This conflict interaction is the one finding that does **not** survive the dynamic-panel (Nickell) bias correction (it collapses to ns) — unsurprisingly, since it is built on **idp_baseline**, which is derived from the lagged displacement level. It should be treated as suggestive only; the food result (M11c) does not share this fragility and is corroborated by the refugee mirror. See *Robustness — dynamic-panel bias*.

M12 — border cercles leak food-stressed populations

Motivation. Cercles on Mali's borders share a land crossing with neighbouring countries. A food-stressed population near a border has an exit option — cross-border refugee flight — that an interior cercle does not. The food effect may therefore be smaller (or even opposite) in border cercles.

Moderator. **is_border** = 1 for cercles with cumulative refugee outflow > 1,000 (20 cercles identified from UNHCR data).

Interaction	coef	p	ΔR^2
food $\times$ is_border	−0.00478	0.034**	+0.002
spi6_13_18m $\times$ is_border	+0.039	0.365 ns	−0.001

Interaction	coef	p	ΔR^2
fatalities_7_9m × is_border	−0.019	0.685 ns	0.000
W-fatalities_10_12m × is_border	−0.073	0.290 ns	+0.001

Only the food channel is dampened in border cercles. This is **not wave absorption** — it is **leakage**: food-stressed populations near the border substitute cross-border refugee flight for internal IDP registration. The dampened IDP response is not a reduction in total displacement but a reallocation to the refugee channel. This interpretation is confirmed directly in Part IV.

Sensitivity to border threshold. **is_border** is defined as cumulative refugee outflow > 1,000, giving 20 border cercles. Re-running with thresholds of 500 (31 cercles), 2,000 (16 cercles), and 5,000 (11 cercles) gives significant **food × is_border** at ** or *** in all cases, with coefficients ranging from −0.005 to −0.008. The finding is not sensitive to where the threshold is drawn.

M13 — population size buffers climate and food (not conflict)

Motivation. Populous cercles (cities, regional capitals) have more diversified economies and deeper food markets. They may absorb livelihood shocks that would displace people in smaller, more food-dependent cercles.

Moderator. **log(population_2015)**, standardised across cercles.

Interaction	coef	p	ΔR^2
p_anom × log(pop)	−0.00181	0.010***	+0.003
food × log(pop)	−0.00184	0.056*	+0.001
fatalities_7_9m × log(pop)	−0.015	0.463 ns	0.000
W-fatalities_10_12m × log(pop)	−0.037	0.327 ns	0.000

The interaction coefficients tell a precise story about how population moderates each channel.

Acute rainfall (p_anom × pop, −0.00181*):** In small cercles (−1 SD population), the rainfall slope is **+0.00075** — rainfall *increases* displacement. In large cercles (+1 SD), the slope is **−0.00287** — rainfall is strongly protective. The sign actually *flips* across the population distribution. Small, rural cercles are so dependent on rain-fed agriculture that a rainfall anomaly directly affects livelihoods and can push people to move. Populous cercles have diversified non-agricultural income; rainfall variation doesn't threaten livelihoods the same way.

Food prices (food × pop, −0.00184):* The slope stays positive at both ends — food stress increases displacement everywhere — but the magnitude halves: +0.00731 in small cercles vs +0.00363 in large cercles. Populous cercles have deeper markets, more wage employment, and greater ability to substitute away from food price shocks. They don't eliminate the food-displacement relationship; they dampen it by roughly half.

Conflict (both null): Violence displaces uniformly regardless of city or cercle size — there is no economic buffer against security threats.

The results are fully robust to population year: re-running with 2019 population gives identical coefficients and p-values to four decimal places, as the cross-cercle population ranking is perfectly stable between 2015 and 2019 (corr = 1.000).

M14 — cities buffer climate and food (not conflict)

Motivation. Major cities may differ from border or high-population cercles through institutional capacity — market integration, social services, employment diversity. A discrete city dummy tests whether urban status buffers shocks over and above the continuous population effect in M13.

Moderator. **is_urban** = 1 for Bamako and 5 regional capitals (Kayes, Ségou, Sikasso, Mopti, Gao) — 6 cercles.

Interaction	coef	p	ΔR^2
streamflow × is_urban	+0.000	0.970 ns	0.000
spi6_13_18m × is_urban	−0.087	0.040 **	−0.003
spi6_flood_dur × is_urban	−0.033	0.328 ns	−0.003
p_anom × is_urban	−0.002	0.227 ns	+0.002
food × is_urban	−0.00798	0.028 **	−0.001
fatalities_7_9m × is_urban	−0.030	0.248 ns	0.000
W-fatalities_10_12m × is_urban	−0.034	0.586 ns	0.000

Both food and the slow SPI channel are significantly buffered in cities (both **). The conflict channel is null across all M5 drivers. Cities dampen livelihood-driven displacement through market depth and employment alternatives; they do not change the violence-displacement relationship.

On the negative ΔR^2 . Both significant interactions show $\Delta R^2 < 0$ — R^2_{within} *decreases* when the interaction is added. This is paradoxical only if R^2 is interpreted as a simple goodness-of-fit measure. In a two-way FE panel, **rsquared_within** is the fraction of within-unit variance explained *after* demeaning. Adding a new regressor can mechanically reduce this metric if the new term absorbs variance that the entity FE was previously accounting for through the demeaned mean — a known property of PanelOLS, not a sign that the term is harmful. The interaction coefficient being significant ($p < 0.05$) and mechanistically coherent is the correct criterion; the negative ΔR^2 is a measurement artefact of the FE framework, not evidence against the finding.

Sensitivity to urban definition. We test eight alternative definitions to identify the mechanism behind M14's findings:

Definition	n	Mopti/Gao?	food×urban	spi6×urban	W·fat×urban
Bamako + Mopti + Gao	3	✓	−0.010**	ns	ns
Kayes + Ségou + Sikasso	3	×	ns	ns	ns
All 6 named capitals	6	✓	−0.008**	−0.087**	ns
Top-6 by population	6	×	ns	ns	ns
Top-10 by population	10	✓	ns	ns	ns
Top-6 by density	6	×	ns	ns	ns
Top-10 by density	10	✓	ns	ns	ns

Four conclusions emerge. First, **W-fatalities is null across every single definition** — conflict displaces uniformly regardless of how urban status is defined. This is the most robust result in M14: there is no urban buffer against violence under any reasonable definition.

Second, **Kayes, Ségou and Sikasso — despite being larger (population ranks 3–5) and denser than Mopti and Gao — contribute nothing alone**, ruling out population size and density as the mechanism for the food and SPI effects.

Third, **simply including Mopti and Gao in a larger set is not sufficient**: top-10 definitions that include both produce null results, because diluting them with 7–8 non-capital cercles removes the signal.

Fourth, the food and SPI effects are specific to the **6-capital administrative network** as a coherent system. Gao (density rank 34, population rank 12) and Mopti (population rank 10) are not large or dense cities. What they share with Bamako is their role as **regional administrative and commercial hubs** for Mali's most food-insecure and climate-exposed populations. The buffering reflects institutional access — markets, government services, NGO presence — not demographic scale.

M15/M16 — Do named disasters amplify the climate-displacement response?

Motivation. The core question is whether officially declared flood or drought events amplify the displacement response to climate variables — does the SPI or streamflow slope steepen during a named disaster period? If so, the binary event label captures severity information that the continuous physical variables miss.

Data. 20 EMDAT flood + drought events in Mali 2007–2024, spatially assigned to cercles by admin level (admin2 directly; admin1 fanned out to all cercles in the region; admin0 nationally). `flood_active` = 1 for any observation interval overlapping an active flood event for that cercle.

Before testing interactions, we first verify that `flood_active` carries any independent displacement signal at all (M15). If the dummy is null as a main effect, there is little motivation to test whether it amplifies slopes.

M15 — M5 plus event dummy as main effect:

$$y_{it} = [M5] + \gamma \cdot \text{flood_active}_{it} + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Spec	coef	p	R²_within	ΔR²
M5 baseline	—	—	0.0992	—
M15a: M5 + flood_active	+0.153	0.038**	0.0983	−0.001
M15b: M5 + drought_active	−0.015	0.788 ns	0.0990	0.000

`flood_active` is significant (+0.153**, p=0.038): named floods carry a displacement signal beyond the continuous physical intensity. `drought_active` is null — the continuous channels fully encode drought severity. The negative ΔR² for M15a is the FE non-monotonicity artefact. All M5 coefficients remain stable (W-fatalities changes by < 1%, SPI channels by < 2%). This confirms named flood events are worth testing as slope amplifiers.

M16 — does the named event steepen the climate slope? We add an interaction between each continuous climate variable and the event dummy:

$$y_{it} = [M5] + \gamma \cdot \text{flood_active}_{it} + \delta \cdot (f(x_{it}) \times \text{flood_active}_{it}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Spec	interaction	coef	p
M16a	streamflow × flood	+0.00012	0.277 ns
M16b	p_anom × flood	+0.00055	0.756 ns
M16c	spi6 × flood	+0.113	0.074 *
M16d	spi6_flood_dur × flood	+0.063	0.086 *
M16e	spi6 × drought	−0.035	0.781 ns
M16f	p_anom × drought	−0.006	0.100 ns
M16g joint	all 6 together	—	0/6 sig

M16c and M16d are marginally significant in isolation, but **0/6 survive the joint test**. The amplifier hypothesis fails: during named flood events, the displacement slope on SPI and streamflow does not systematically steepen. The continuous variables capture the relevant physical intensity; the named-event label adds a level shift (M15) but not a slope change (M16).

M17 — unified stacked model

M7–M16 each tested one extension in isolation. M17 stacks the four validated extensions into one specification to check whether they survive jointly and to measure their combined explanatory power.

The four extensions in M17 — each from a validated standalone test:

- **food × idp_baseline (M11c)** — food dampened in high-displacement cercles (cross-border substitution; Part IV)
- **W·fat × idp_baseline (M11_W)** — conflict spillover also dampened
- **I(streamflow > p75) (M7d)** — current-window flood threshold (added alongside the continuous streamflow already in M5)
- **p_anom × log(pop) (M13)** — acute rainfall buffered in populous cercles

$$y_{it} = [M5] + \gamma_1(f(\text{food}) \cdot z_{idp}) + \gamma_2(f(W \cdot \text{fat}) \cdot z_{idp}) + \gamma_3 \cdot \mathbf{1}[\text{stream} > p_{75}] + \gamma_4(f(p_anom) \cdot z_{pop}) + \delta' C_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Results:

Term	coef	p	
food × idp_base	−0.00501	0.004	***
W·fat × idp_base	−0.14461	0.003	***
p_anom × pop	−0.00161	0.018	**
I(stream > p75)	+0.08526	0.101	ns

Three of the four survive jointly. The streamflow threshold, marginally significant standalone (M7d, +0.099*), weakens to ns when stacked — its upper-tail signal overlaps with the other terms once all are in the model. The three heterogeneity interactions (M11c, M11_W, M13) remain strongly significant. R^2_{within} rises from **0.099 to 0.112** (+0.013).

M17 (stacked model) — standardized coefficients with 95% CI
(faded = ns; † = not robust to dynamic-panel correction)

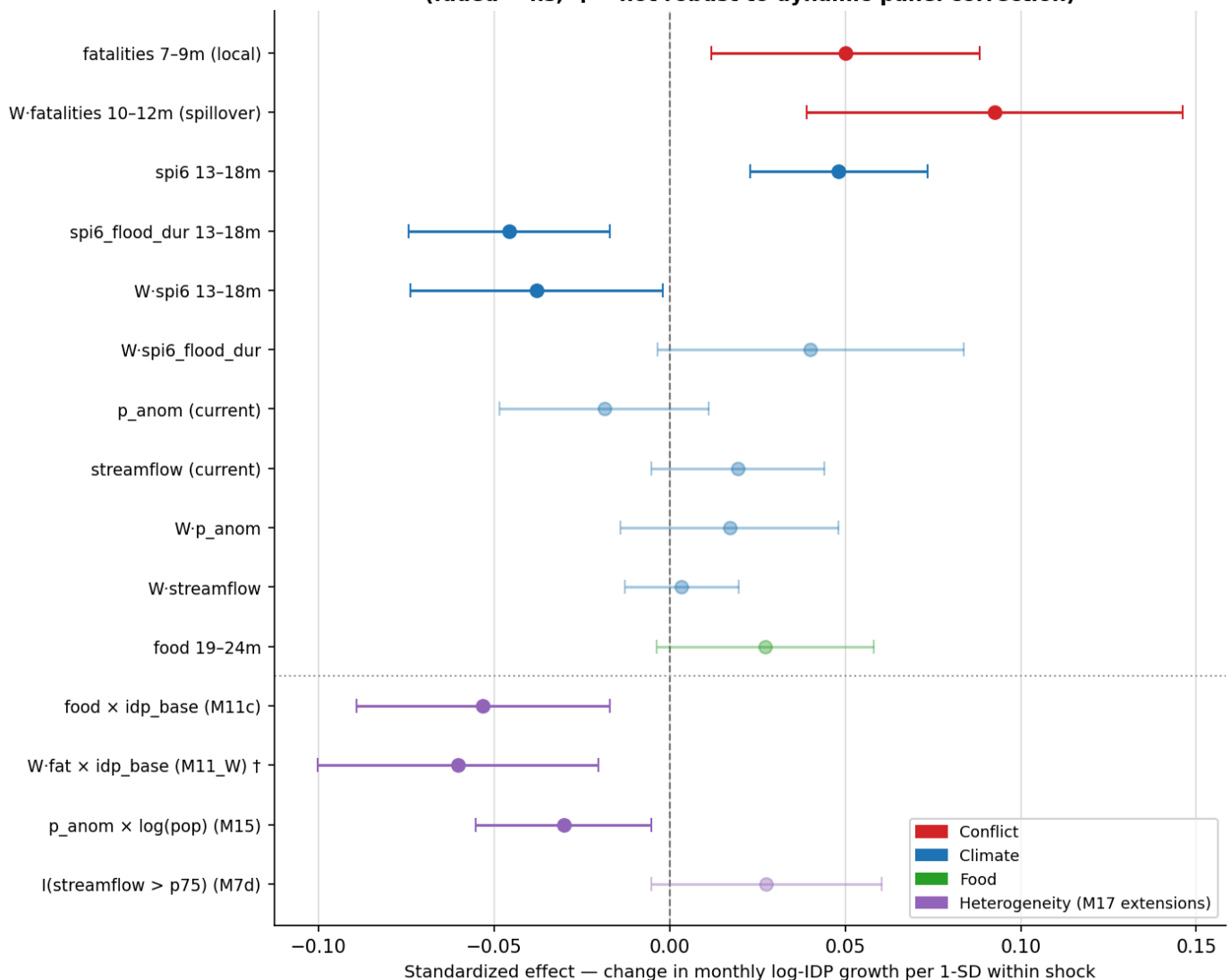


Figure 6. M17 (stacked model) coefficients as **standardized effects** (change in monthly log-IDP growth per 1-SD within-cercle shock), with 95% CIs; faded = not significant. Above the line are the M5 main effects; below are the four heterogeneity extensions (M11c, M11_W, M13, M7d threshold). The acute-water main

effects (`p_anom`, `streamflow`) lose *individual* significance here because their signal is partly absorbed by the `p_anom × log(pop)` interaction and the streamflow threshold. `W·fat × idp_base` does not survive the dynamic-panel correction (see *Robustness*).

A notable side effect: the `W·fatalities` main effect **strengthens from +0.252 to +0.373** in M17. M5's coefficient was an average across the IDP-history distribution; M17 separates the absorption component (the `W·fat × idp_baseline` interaction) from the baseline conflict effect, so the main coefficient now represents the conflict effect in *low*-IDP-history cercles where this dampening is minimal.

Model progression. Before decomposing M17, here is how far the modelling has come:

Model	R ² _within	Δ
Controls + FE only	0.048	—
M5 (selected drivers)	0.099	+0.051
M17 (+ heterogeneity)	0.112	+0.013

The Stage A–E funnel more than doubled within-R² over the controls-only baseline (+0.051); the RF-guided functional-form tests and heterogeneity extensions (M6–M14) added a further +0.013. The LOO below shows which families drive M17's R²=0.112.

LOO block decomposition (comparable basis). Each block's share of its model's driver contribution — the R²_within it gains over the AR-control baseline (0.048) — placed side by side with the M5 decomposition:

Driver block	M5	M17
Conflict (local + spillover; + <code>W·fat×idp</code> in M17)	72%	83%
Climate (8 water/SPI terms; + threshold & <code>p_anom×pop</code> in M17)	16%	8%
Food (+ <code>food×idp</code> in M17)	~0% (null)	6%

Conflict dominates both models — its share *rises* from 72% to 83% once the `W·fat × idp_baseline` interaction joins the conflict block. Food climbs from null to 6%, the `food × idp_baseline` interaction unlocking signal M5 was averaging away. Within the climate block, the slow SPI terms contribute essentially nothing once stacked; the acute water terms carry it.

Temporal stability — the 2020 structural break

Mali's panel spans a major political rupture: the **military coup of 18 August 2020**, after which the international counter-insurgency presence wound down and the conflict shifted toward a decentralised rural insurgency. Splitting the sample at 2020-07 (just before the coup) reveals that the displacement regime does not merely weaken — it **inverts**.

M5 channel	Pre-2020-07	Post-2020-07
n	1,111	306
R ² _within	0.104	0.365
local conflict (fat 7–9m)	+0.051 ns	+0.044 ***
conflict spillover (W·fat 10–12m)	+0.264 **	–0.110 ** (collapses; ns after bias correction)
SPI flood (spi6 13–18m)	+0.150 ***	+0.044 ns
flood duration	–0.065 ***	–0.020 ns
current rainfall (p_anom)	–0.0026 **	+0.0002 ns
streamflow	+0.00011 **	–0.00001 ns
food (19–24m)	+0.014 **	+0.001 ns

Three things change at once. First, **the model fits far better post-coup** (R²_within 0.10 → 0.37): displacement became much more predictable, because a single factor came to dominate. Second, **that factor is local conflict** — `fatalities_7_9m`, non-significant before, becomes the strongest driver after. Third, and most striking, **the conflict spillover collapses** — the robustly positive pre-2020 effect (+0.26) vanishes, turning to –0.11 in FE (and to a non-significant –0.06 under the dynamic-panel correction; see *Robustness*). Violence in neighbouring cercles no longer pulls people in as IDP arrivals.

The coherent explanation is that **post-coup road blockages broke the spatial flight mechanism**. Before 2020, people fled conflict by moving toward safer neighbouring cercles, registering as arrivals there — the positive spillover. After 2020, with roads cut and movement restricted, violence displaces people *in place* and cross-cercle flight shuts off — which is exactly why the spillover channel **vanishes** (a switched-off reception channel implies a *null* spillover; the FE estimate even turns negative, but the bias-corrected estimate is indistinguishable from zero). Every climate and food channel — which operate on slower, livelihood-mediated timescales — is swamped by the violence and goes null.

Is 2020 really the break? Re-splitting at every half-year from 2018 to 2021 confirms the break is genuinely around the coup, not an artefact of the chosen cut:

Split point	n_post	R ² _post	spillover (post)
2018-07	620	0.15	–0.049 ns
2019-07	445	0.19	–0.090 ns
2020-07	306	0.37	–0.110 **

Split point	n_post	R ² _post	spillover (post)
2021-07	219	0.42	−0.143 **

Two signatures pin the break to 2020. First, the post-period R^2 jumps sharply — about 0.15 for pre-2020 cuts, but 0.31–0.42 once the cut reaches 2020. Second, the collapse of the positive spillover is only *significant* from 2020 onward: cutting at 2018 or 2019 gives a weak, non-significant negative spillover (those post-windows are still mostly pre-coup), but as the cut moves into the post-coup period the negative spillover strengthens monotonically (−0.09** → −0.14***, in FE). The structural break is a 2020 phenomenon.

All magnitudes in this report describe 2014–2020 Mali. The post-2020 regime is a different system, and the static road-distance spatial weights cannot represent blocked roads; forecasting the current period would require a road-state-aware weight matrix and re-estimation on a recent subsample.

Part IV — External validation: refugee outflow

DTM IDP data and UNHCR refugee data are collected by different agencies, using different methodologies, recording different physical outcomes. If the same driver families predict refugee outflow with coherent signs, that is strong cross-outcome validation.

The refugee outcome is each cercle's outflow of Malian refugees **across the border to neighbouring countries**. Because this is movement *out of Mali*, the spatial-spillover terms — which capture redistribution *between* Malian cercles — do not apply: cross-border flight is driven by the cercle's *own* conditions pushing people over the international border, not by neighbour-weighted conditions inside Mali. The refugee model therefore uses local drivers only.

Outcome and specification. The refugee outcome is **log1p** of the **average monthly outflow** over the same survey interval (**t1**, **t2**) used for IDP — a flow level, where the IDP outcome is a per-month growth rate of a stock; both are log-scale per-month quantities aligned to identical intervals ($n = 1,417$). The specification keeps two-way FE, clustered SE, and the same controls, with **six local drivers**: five are M5's local terms (SPI flood and duration, current rainfall and streamflow, food), and conflict enters at a refugee-tuned **1–3m lag** — cross-border flight responds to *recent local* violence directly, with none of the registration delay that gives IDP its 7–12m conflict lags. Because the outcomes are on different scales (flow level vs growth rate), the mirror ratios below indicate equal-and-opposite *responses*, not a person-for-person accounting identity.

The refugee data is sparse and geographically concentrated. Unlike internal displacement, which is spread across the country, cross-border outflow is overwhelmingly a **northern** phenomenon: ~80% originates in the three northern regions (Tombouctou 41%, Taoudénit 23%, Gao 15%), and just **three cercles account for ~half** of all outflow (the top six ≈ 74%). It traces to the post-2012 occupation of the north, with refugees fleeing chiefly to **Mauritania, Niger, and Burkina Faso**, and the series runs longer than the IDP panel (2012–2025). The refugee model is therefore identified off a smaller, northern base than the broad-coverage IDP model — which makes the cross-outcome agreement below (especially the mirror substitution) a check across genuinely *different* geographies, but also means the refugee findings lean on a handful of northern cercles and should be read as corroboration rather than independent proof.

The refugee outcome is well explained by the same drivers ($R^2_{\text{within}} = 0.119$), but with a different driver mix. Comparing each driver on both outcomes under a common local specification (recent conflict at 1–3m):

Driver	refugee	IDP
local conflict (1–3m)	−0.016 ns	+0.039 *
SPI flood (13–18m)	+0.175 *	+0.115 ***
flood duration (13–18m)	−0.059 ns	−0.047 ***
current rainfall (p_anom)	+0.0052 ***	−0.0020 **
streamflow	+0.000 ns	+0.0001 **
food (19–24m)	+0.016 *	+0.0065 *

Two contrasts stand out. **Refugee outflow is driven by climate and food, not conflict** — local conflict is non-significant for refugees, whereas it predicts IDP (and far more strongly through neighbour *spillover*: the M5 structure reaches 0.099 vs 0.058 for this local spec). And the climate channels cross-validate by sign: **SPI flood carries the same positive sign on both outcomes** (+0.175* refugee, +0.115*** IDP), while **current rainfall flips sign** (+0.0052*** refugee vs −0.0020** IDP) — protective for inland livelihoods (less IDP) but causing Niger River flooding that drives refugees northward. Food carries the same positive sign on both.

Why food appears to dampen displacement. Part III left a puzzle: food-price shocks seemed to *reduce* internal displacement in cercles with a long displacement history (M11c) and those near a border (M12). Why would food stress reduce flight? The refugee data answers it. The same two interactions, run on refugee outflow, come out **equal and opposite** to the IDP versions:

Interaction	refugee	IDP	ratio
food × idp_baseline (M11c)	+0.0057***	−0.0054***	−1.05
food × is_border (M12)	+0.0080**	−0.0050**	−1.61
p_anom × log(pop) (M13)	ns	−0.0016**	IDP-specific
food × is_urban (M14)	ns	−0.0081**	IDP-specific

In these cercles a food-price shock does not dampen displacement — it **redirects** it, shifting people from internal IDP registration to cross-border refugee flight (near 1:1 for displacement-history cercles, larger still for border cercles). Food stress is a *substitution between displacement channels*, not a reduction in displacement. The other two interactions (M13, M14) have no refugee counterpart — they are genuinely IDP-specific.

Robustness — dynamic-panel (Nickell) bias

M5 conditions on the lagged displacement level (`log1p_idp_t1`) to absorb mean reversion. In a two-way fixed-effects panel, including a lagged dependent level induces **Nickell bias**, which can distort the driver coefficients. We test this with an instrumental-variables correction.

The instrument. The natural Anderson–Hsiao instrument is the previous survey's level, but it is *invalid here*: because the outcome is a **change** over consecutive intervals that share an endpoint, the residuals carry negative serial correlation ($AR(1) \approx -0.23$). We therefore use the **lag-2 level** as the instrument (the Arellano–Bond remedy under MA(1) errors); its first stage is strong.

Quantity	FE	IV (lag-2)	verdict
M5 conflict spillover	+0.251***	+0.208***	robust (−17%)
food × idp_base (M11c)	−0.0050***	−0.0030**	robust
p_anom × log(pop) (M13)	−0.0016**	−0.0018***	robust
l(stream > p75)	+0.085 ns	+0.080 ns	stable
W·fat × idp_base (M11_W)	−0.145***	−0.020 ns	not robust
spillover, pre-2020	+0.264**	+0.241***	robust
spillover, post-2020	−0.110**	−0.062 ns	sign holds, sig. weakens

Three conclusions:

1. **The headline results are robust.** The conflict spillover (M5), the food-substitution interaction (M11c), and the population buffer (M13) all survive the correction with the same signs and significance; the bias shaves only ~17% off the conflict spillover.
2. **The structural break holds in direction.** The spillover is positive pre-2020 (+0.241***) and negative post-2020 (−0.062); the reversal survives the correction, though the post-coup estimate loses significance in the smaller IV sample (n≈300).
3. **One exception — the conflict wave-absorption interaction (W·fat × idp_base, M11_W) is not robust:** it collapses to non-significance. This is expected — it is built on `idp_base_line`, which is mechanically derived from the lagged level, making it the term most entangled with the dynamic-panel bias. The *food* substitution result (M11c) does not share this fragility and is independently corroborated by the refugee mirror (Part IV).

Robustness — dropping the climate spillover terms

Only one of the four climate spillover terms is significant (`W·spi6_13_18m`, −0.311**); the other three (`W·spi6_flood_dur`, `W·p_anom`, `W·streamflow`) are null. A natural question is whether the conclusions depend on these weak neighbour-climate terms at all. We refit both M5 and M17 with all four climate spillovers removed.

Quantity	M5 full	M5 no-CS	M17 full	M17 no-CS
R ² _within	0.099	0.095	0.112	0.110
conflict (local + spillover)	—	unchanged	—	unchanged
food / M11c / M11_W / M13	—	unchanged	—	unchanged
<code>spi6_13_18m</code> (local flood)	+0.139	+0.107	+0.121	+0.091
<code>spi6_flood_dur</code>	−0.054	−0.044	−0.042	−0.031

Two conclusions. **First, every headline result is robust:** conflict dominance, the food substitution interaction (M11c), the conflict wave-absorption interaction (M11_W), and the population buffer (M13) are all unchanged in sign and significance; the fit barely moves (−0.005 in M5, −0.002 in M17). The displacement story does not rest on the neighbour-climate terms.

Second, the climate spillovers cannot simply be deleted as noise. The local SPI term and its spillover are an *amplifying pair*: within-cercle they correlate +0.42 with **opposite** signs (local +0.139, spillover −0.311), which by the suppression/amplification rule means each inflates the other. Dropping the spillover therefore shrinks the local SPI coefficient by ~23% (+0.139 → +0.107 in M5; +0.121 → +0.091 in M17). The neighbour-flood signal is real and coupled to the local one, not redundant — the moderate cross-cercle correlation of rainfall (`p_anom` +0.57, `spi6` +0.42) is enough to entangle the pair but not enough to make the spillover disposable. We retain the full spillover set so the local SPI coefficient is not artificially deflated.

Robustness — out-of-time validation

All cross-validation elsewhere in this report is **entity-blocked** (held-out *cercles*). A reviewer rightly asks whether the model also generalises across *time*. We re-ran the M5 out-of-sample test holding out **time** instead of *cercles* (RF on within-cercle–transformed M5 features, entity means computed on the training fold only):

Validation scheme	OOS R ²
Held-out cercles (entity-blocked, same pipeline)	+0.070
Held-out time (contiguous 5-fold)	−0.24
Forward-chaining (train past → test future)	−0.40
Train pre-2020 → test post-2020	−1.05
Within pre-2020: early → late	−0.32

The model **generalises across space but not across time**: every time-blocked scheme yields a *negative* OOS R^2 (worse than predicting the mean). This is **not solely the 2020 coup** — the within-pre-2020 early→late split is also negative, so the driver→displacement mapping is **non-stationary throughout the panel**. Mechanically, a substantial part of the in-sample within- R^2 rests on the *time* fixed effects (common annual/seasonal shocks), which by construction carry no out-of-time predictive content; the drivers alone do not predict the time path.

Implication. The findings should be read as explaining **within-period, cross-cercle variation** — *why some cercles displace more than others in a given period* — not as a forecast of *when* displacement will rise. Operational forecasting would require re-estimation on a recent window and a non-stationarity-aware design. (Reproducible via [scripts/run_temporal_cv.py](#) and [scripts/run_m5_bootstrap.py](#).)

Robustness — post-selection inference and model stability

The funnel searches a large space (up to 3.9M specifications), so the final **p-values are post-selection** — the same data selected *and* tested the model, biasing them toward significance. They should be read as **descriptive, not confirmatory**. Confirmation instead rests on four out-of-selection checks:

- 1. **Selection stability** (cluster bootstrap, 150 resamples, *re-running the selection each time*). The backbone is robustly re-selected: conflict anchor = **fatalities 84%**, spillover lag = 10–12m **89%**, channel-1 = SPI (top both-significant pair) **79%**. The finer choices are not: the local conflict lag is 7–9m in only **40%** (4–6m 31%, 1–3m 18%), and the second climate channel is a coin-flip — but M5's acute-water terms are significant in 36% of resamples vs **8%** for M4's temperature, which justifies the M5-over-M4 choice.
- 2. **Pre-specified replication** (no search). Three parsimonious, theory-lagged models (conflict local + spillover at a round lag, one climate term, one food term) all reproduce the headline — conflict spillover **+0.28 to +0.32*****, dwarfing local conflict (+0.04**), with **food null** throughout; conflict accounts for essentially all of the driver contribution. The ranking is not an artifact of the search.
- 3. **External replication**. The independent refugee outcome (Part IV) reproduces the climate and food patterns.
- 4. **Out-of-time validation** (previous section) — done, and honestly negative.

Reading. Treat the funnel as exploratory *selection*; the **backbone** (conflict spillover, SPI flood, food-null) is confirmed by stability + pre-specification + external replication, while the **finer** selections (exact local lag, second climate channel) are correctly held as suggestive. (Reproducible via [scripts/run_selection_bootstrap.py](#) and [scripts/run_prespecified.py](#).)

Conclusion — what we found

What drives displacement:

- **Violence is the biggest driver — and it spreads.** Violence in *neighbouring* communities pushes people to flee even more than violence at home — about five times more.
- **Climate matters, but less — and both extremes push people out.** Severe flooding about a year earlier, and sudden rainfall or river surges in the moment, drive displacement — and so does drought, on a longer horizon (about two years). All are secondary to violence, but real.
- **Food prices don't push people out — they redirect them.** This shows up most clearly in two kinds of places: communities with a long displacement history, and **those on a border with a ready cross-border exit**. In both, food-price spikes send people **across the border as refugees** rather than into internal displacement — the same number flee, but through a different channel.

Who is affected depends on local conditions:

- The same shock lands differently depending on a community's history and size. Places with a long displacement history, and larger towns and cities, **absorb or redirect** food and climate shocks — but **not violence, which displaces people everywhere regardless**.
- **Yet when we combine every driver and every one of these differences into a single model, the verdict is unchanged: conflict alone accounts for about half of everything we can explain; climate and food make up the smaller remainder.**

How much this explains, and when:

- The machine-learning cross-check confirmed the findings and added a sobering note: **displacement is genuinely hard to predict**. Much of why people flee comes down to local factors no dataset captures — community ties, leadership, household decisions.
- An independent refugee dataset — different agency, different outcome — reproduced the climate and food patterns, which strengthens confidence that these are real signals.
- **On causality:** the strongest causal reading is for **climate** (weather is exogenous) and **conflict spillover** (neighbour violence can't be caused by local displacement); *local* conflict and food are weaker and should be read as credible **associations**, not proven effects.
- The main results also survive a **dynamic-panel robustness check** (correcting for how the model carries past displacement forward): the conflict, climate, and food-redirection findings all hold. The one exception is a *secondary* pattern — the dampening of conflict spillover in already-displaced areas — which does not survive and should be treated with caution.
- These patterns describe **Mali up to 2020**. After the August 2020 coup the whole pattern shifts: local violence becomes the dominant driver, and the cross-community spillover **collapses** — blocked roads appear to cut off flight to neighbours, so the channel that pulled people into safer cercles switches off (the effect drops from strongly positive to indistinguishable from zero), while the climate and food signals fade. The model explains the past well; the post-coup period is a different system.

The findings in detail:

Driver	finding	robustness
Conflict spillover	Dominant. Violence in neighbouring cercles (10–12m lag) is the single largest predictor. Spillover 5x local.	100% of 987k models; stable across conflict source, spatial weights, SE specification
SPI flood channel	Real but secondary. Wet anomaly 13–18m ago → displacement; prolonged prior flooding → wave absorbed.	272/5,880 both-significant pairs; RF confirms linear, not threshold

Driver	finding	robustness
Acute water shock	Real and contemporaneous. Current rainfall protective; active streamflow displacing.	Threshold confirmed (M7d); robust across lag specs
Food prices	Null on average. Redirects rather than suppresses — food shocks shift displacement from internal (IDP) to cross-border (refugee).	All 21 food variables null in multivariate; M11c/M12 mirror in refugee data ***
Food spillover (W-food)	Not informative. Dropped — $p=0.729$; food price is a near-national time series.	Confirmed across all 21 food variables

Limitations and open questions

Identification — graded, not uniform. We have no instrument or natural experiment for the drivers themselves (the dynamic-panel correction above instruments only the lagged *outcome*, not conflict, climate, or food), so these are not clean causal point estimates. The design nonetheless supports a graded causal reading: two-way FE removes all time-invariant confounders and common shocks; drivers enter only at lags (local 7–9m, spillover 10–12m for conflict; 13–24m for climate) with the contemporaneous window excluded, making reverse causality implausible; **climate** is plausibly exogenous to displacement (weather is not caused by flight); and the headline **conflict-spillover** effect cannot be reverse-caused by local IDP counts (*W·fatalities* is hard to argue is produced by local arrivals). The credible-causal claim is **strongest for climate and conflict-spillover**, weaker for *local* conflict (not exogenous) and *food* (little within-cercle variation), and is bounded throughout by remaining **time-varying** cercle-specific confounders, which FE does not remove.

Net stock changes, not gross flows. The outcome is the change in the registered IDP stock ($\Delta \log_{1p}(\text{IDP})$) — the **net** of arrivals – departures – returns – attrition over each interval. It cannot decompose gross inflows from outflows: a small net change can mask large offsetting flows, and "arrivals/departures" language (e.g. for the conflict spillover) is **mechanism inference, not direct measurement**. Three things mitigate this: the net stock is itself the policy-relevant displacement *burden*; the local/spillover split is a partial directional proxy (the positive spillover is most plausibly *arrivals*, since neighbour violence has no channel to reduce the focal cercle's own departures); and Part IV's refugee outflow is a genuine **gross flow** that cross-validates the climate/food patterns. Returns-home and mortality are nonetheless folded into the net change and cannot be separated.

Edge-of-data claims. The panel is thin ($n = 1,417$; 44 cercles; ~65% of raw changes zero), and some analyses push to its limits — the 2020 structural break (post-period $n = 306$), the fine heterogeneity interactions, the three-way SPI decomposition, and a 3.9M-model search (multiple comparisons). The **headline trio is robust** — conflict dominance, climate secondary, food-redirect — supported by out-of-sample RF cross-validation (the M5-restricted forest generalises *better* than the full-feature one) and by parsimony chosen deliberately *because* the data is thin. The **finer claims** (post-2020 regime, detailed heterogeneity, SPI sub-facets) are **suggestive, edge-of-data** results — consistent with the graded-confidence framing and the explicitly low predictability ($R^2 \approx 0.10\text{--}0.14$).

Out-of-time non-generalization. The model is validated on held-out *cercles*, not held-out *time*. A direct out-of-time test (forward-chaining and pre/post-2020 hold-outs) yields **negative** OOS R^2 — *including within the pre-2020 regime* — so the driver→displacement relationship is **non-stationary** and the model does not forecast the time path. The results explain cross-cercle variation within a period, not period-to-period movements; forecasting would require re-estimation on a recent window. See *Robustness — out-of-time validation*.

Post-selection inference. The funnel's p-values are computed on the same data used to select the model and are therefore **not valid confirmatory p-values** (selective inference / winner's curse). We treat selection as exploratory; the backbone is confirmed *out of selection* — bootstrap selection stability (re-selected 79–89%), a pre-specified no-search replication, and the refugee external check — not by the post-selection p-values, which should not be over-interpreted. See *Robustness — post-selection inference and model stability*.

R² and causal importance. R^2_{within} is the primary fit criterion for model selection, applied alongside the p-value and coefficient-sign criteria. It is imperfect — an *in-sample* measure, and non-monotonic under the two-way FE transform — but it is the natural, interpretable metric in the within-cercle space, and the selections were cross-checked against out-of-sample R^2 (RF cross-validation). Crucially, R^2_{within} measures explanatory fit, not causal effect size: the conflict > climate > food ranking reflects relative *explanatory* importance, not a direct measure of causal contribution.

Non-linearity scan. The headline non-linearity screen (Part II) focuses on the M5 features, but the scan was also extended to **all 282 candidates**, crossed with held-out importance: it surfaces no new driver — the only significant hit re-discovers the documented drought channel through its duration facet. Purely non-linear effects below that scan's noise floor could still exist, but the full-feature RF's out-of-sample performance bounds any such signal to be small.

Two-lag SPI. The slow flood signal (spi6 13–18m) and a longer-horizon drought signal (spi6 19–24m) are each individually significant (drought -0.064^{**} , $p=0.015$) and nearly orthogonal in the within-variation (within-correlation 0.03). Adding the drought term alongside the flood term leaves every other M5 coefficient essentially unchanged but *lowers* R^2_{within} ($0.099 \rightarrow 0.094$) — the fixed-effects non-monotonicity discussed earlier — so M5 keeps only the flood signal for parsimony. The drought channel is real (it appears in Stage A and is examined in M6) but is not in the final model; see *A second drought channel* in Part II.

Post-2020 regime change. All magnitudes in this report describe 2014–2020 Mali. After the August 2020 coup the displacement regime shifts — local conflict becomes dominant and the positive cross-cercle spillover collapses (see *Temporal stability — the 2020 structural break*). The spatial weight matrix assumes static road connectivity, which post-coup road blockages violate; forecasting the current period would require a road-state-aware W and re-estimation on a recent subsample.